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Introduction

The fast expansion of urban mobility services has encouraged data-driven approaches to better understanding transportation usage patterns. One such project is the Bike Sharing System, which provides a sustainable and efficient mode of transportation in metropolitan areas. This study makes use of the Bike Sharing Dataset <https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset>, offered by the UCI Machine Learning Repository (Fanaee-T & Gama, 2014), which contains daily and hourly bike rental data collected by Capital Bikeshare in Washington, D.C. over two years (2011-2012).

The dataset includes various variables that impact bike demand, such as weather conditions, seasonal factors, day types (working days and holidays), and temporal indicators like year and month. The primary goal of this investigation is to develop and test several linear regression models to predict the total bike rental count (cnt) using specific explanatory factors.

The data were initially collected by the Capital Bikeshare system through automatic log entries at bike rental stations, which documented the number of casual and registered users every hour and day. Weather-related indicators were gathered from the National Oceanic and Atmospheric Administration (NOAA) and compared to time-stamped rental data.

By performing this regression analysis, the study hopes to find the most significant predictors of bike usage and develop a robust model that can apply well to previously unreported data, allowing for better planning and demand forecasting for bike-sharing operations.

Methodology

Dataset Overview

This project uses the Bike Sharing Dataset, there are 13 features in this dataset, including continuous, integer, categorical, binary, and date data types.

Target Feature:

- cnt: Total count of bikes rented (combined casual and registered users)

Predictor Variables:

- instant: record index
- dteday : date
- season : season (1:springer, 2:summer, 3:fall, 4:winter)
- yr : year (0: 2011, 1:2012)
- mnth : month (1 to 12)
- hr : hour (0 to 23)
- holiday : weather day is holiday or not
- weekday : day of the week
- workingday : if day is neither weekend nor holiday is 1, otherwise is 0.
- weathersit :
 1. Clear, Few clouds, partly cloudy, partly cloudy
 2. Mist + Cloudy, Mist + Broken clouds, Mist + Few clouds, Mist
 3. Light Snow, Light Rain + Thunderstorm + Scattered clouds, Light Rain + Scattered clouds.
 4. Heavy Rain + Ice Pallets + Thunderstorm + Mist, Snow + Fog
- temp : Normalized temperature in Celsius. The values are divided to 41 (max)
- atemp: Normalized feeling temperature in Celsius. The values are divided to 50 (max)
- hum: Normalized humidity. The values are divided to 100 (max)
- windspeed: Normalized wind speed. The values are divided to 67 (max)
- casual: count of casual users
- registered: count of registered users

To avoid redundancy, variables like casual, registered, instant and dteday were removed. Furthermore, atemp was excluded from some models due to its significant collinearity with temp.

Data Preparation and Transformation

- All categorical variables (season, yr, mnth, weekday, holiday, weathersit) were converted to factors.
- The response variable (cnt) was kept in its original scale for interpretability.
- A time index (trend) was added to capture the long-term time pattern in demand.

- A lag variable (cnt_lag1) was created to model short-term autocorrelation.

Model Development

Three distinct modelling approaches were used to get the best linear regression model:

1. **Adjusted R² Subset Selection**
The leaps package's regsubsets() function was used to determine the optimal model based on adjusted R². The selected variables were then manually entered into a simplified model.
2. **Stepwise Regression (AIC-based)**
Stepwise forward and backward selection is performed using the MASS package's stepAIC() function to optimize model fit based on the AIC score.
3. **Time Series–Aware Model**
Given the temporal nature of the data, models were fitted using additional variables to deal with autocorrelation:
 - A linear trend variable (trend)
 - A lagged count variable (cnt_lag1)

Model Evaluation and Diagnostics

All models were examined using the following diagnostics:

- Adjusted R² and AIC to improve model performance.
- Used the Durbin-Watson test and ACF plots to assess residual autocorrelation.
- Normality was assessed using the Shapiro-Wilk test and Q-Q plot.
- Residual and fitted plots for homoscedasticity.
- Determine VIF values for multicollinearity.

Models were evaluated based on residual behaviour and assumption validity, with a final suggestion made based on overall performance.

Result

Data Preparation and Transformation

```
# To clean the R environment
rm(list = ls())

# Loading necessary libraries
library(tidyverse)
library(car)
library(leaps)
install.packages("ResourceSelection")
library(ResourceSelection)
install.packages("pROC")
library(pROC)
library(lmtest)
library(nortest)
install.packages("MASS")
library(MASS)
library(nlme)
```

Figure 1: Importing necessary libraries.

```
# Load the bike dataset
bike ← read.csv("day.csv")
head(bike)
str(bike)
```

Figure 2: Loading the dataset.

```
> str(bike)
'data.frame': 731 obs. of 16 variables:
 $ instant : int 1 2 3 4 5 6 7 8 9 10 ...
 $ dteday : chr "2011-01-01" "2011-01-02" "2011-01-03" "2011-01-04" ...
 $ season : int 1 1 1 1 1 1 1 1 1 1 ...
 $ yr : int 0 0 0 0 0 0 0 0 0 0 ...
 $ mnth : int 1 1 1 1 1 1 1 1 1 1 ...
 $ holiday : int 0 0 0 0 0 0 0 0 0 0 ...
 $ weekday : int 6 0 1 2 3 4 5 6 0 1 ...
 $ workingday: int 0 0 1 1 1 1 1 0 0 1 ...
 $ weathersit: int 2 2 1 1 1 1 2 2 1 1 ...
 $ temp : num 0.344 0.363 0.196 0.2 0.227 ...
 $ atemp : num 0.364 0.354 0.189 0.212 0.229 ...
 $ hum : num 0.806 0.696 0.437 0.59 0.437 ...
 $ windspeed : num 0.16 0.249 0.248 0.16 0.187 ...
 $ casual : int 331 131 120 108 82 88 148 68 54 41 ...
 $ registered: int 654 670 1229 1454 1518 1518 1362 891 768 1280 ...
 $ cnt : int 985 801 1349 1562 1600 1606 1510 959 822 1321 ...
```

Figure 3: Structure of the data.

```

> bike_clean <- subset(bike, select = -c(instant, dteday, casual, registered, workingday))
> # First rename predictors to x1 to x10
> names(bike_clean)[1:10] <- paste0("x", 1:10)
> # Rename the response variable cnt to y
> names(bike_clean)[11] <- "y"
> # Factorize categorical variables (keep numeric levels)
> bike_clean$x1 <- factor(bike_clean$x1) # season
> bike_clean$x2 <- factor(bike_clean$x2) # yr
> bike_clean$x3 <- factor(bike_clean$x3) # mnth
> bike_clean$x4 <- factor(bike_clean$x4) # holiday
> bike_clean$x5 <- factor(bike_clean$x5) # weekday
> bike_clean$x6 <- factor(bike_clean$x6) # weathersit
> # View the clean data
> str(bike_clean)
'data.frame': 731 obs. of 11 variables:
 $ x1 : Factor w/ 4 levels "1","2","3","4": 1 1 1 1 1 1 1 1 1 1 ...
 $ x2 : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
 $ x3 : Factor w/ 12 levels "1","2","3","4",..: 1 1 1 1 1 1 1 1 1 1 ...
 $ x4 : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
 $ x5 : Factor w/ 7 levels "0","1","2","3",..: 7 1 2 3 4 5 6 7 1 2 ...
 $ x6 : Factor w/ 3 levels "1","2","3": 2 2 1 1 1 1 2 2 1 1 ...
 $ x7 : num 0.344 0.363 0.196 0.2 0.227 ...
 $ x8 : num 0.364 0.354 0.189 0.212 0.229 ...
 $ x9 : num 0.806 0.696 0.437 0.59 0.437 ...
 $ x10: num 0.16 0.249 0.248 0.16 0.187 ...
 $ y : int 985 801 1349 1562 1600 1606 1510 959 822 1321 ...

```

Figure 4: Factorised variables.

The dataset has been cleaned and modified to keep only the variables necessary for regression analysis. Redundant or potentially misleading variables such as instant, dteday, casual, registered, and workingday were removed to avoid data leakage and multicollinearity.

Categorical variables such as x1 (season), x2 (yr), x3 (mnth), x4 (holiday), x5 (weekday), and x6 (weathersit) were correctly transformed to factors. This guarantees that R treats them as categorical predictors and generates the necessary dummy variables during model fitting.

The final dataset consists of 731 daily observations and 11 variables. The response variable is y (cnt), and the predictors include both numerical and categorical variables, specifically:

x1: Season

x2: yr

x3: mnth

x4: holiday

x5: weekday

x6: weathersit

x7: temp

x8: atemp

x9: hum

x10: windspeed

y: cnt

Model Development and Evaluation

MODEL 1

To begin, a multiple linear regression model was built utilizing all relevant predictor variables to explain the variation in total bike rental count (cnt). This base model incorporates both numerical and categorical characteristics that could influence daily bike demand. The following formula was used:

$$\text{cnt} \sim \text{season} + \text{yr} + \text{mnth} + \text{holiday} + \text{weekday} + \text{weathersit} + \text{temp} \\ + \text{atemp} + \text{hum} + \text{windspeed}$$

Model Summary

```
> summary(model_full)

Call:
lm(formula = y ~ x1 + x2 + x3 + x4 + x5 + x6 + x7 + x8 + x9 +
    x10, data = bike_clean)

Residuals:
    Min       1Q   Median       3Q      Max
-3944.7  -348.2    63.8   457.4  2912.7
```

Figure 5: Summary of Model 1

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1485.84    239.75    6.198 9.77e-10 ***
x12           884.71    179.49    4.929 1.03e-06 ***
x13           832.70    213.13    3.907 0.000102 ***
x14          1575.35    181.00    8.704 < 2e-16 ***
x21          2019.74     58.22   34.691 < 2e-16 ***
x32           131.03    143.78    0.911 0.362443
x33           542.83    165.43    3.281 0.001085 **
x34           451.17    247.57    1.822 0.068820 .
x35           735.51    267.63    2.748 0.006145 **
x36           515.40    282.41    1.825 0.068423 .
x37            30.80    313.82    0.098 0.921854
x38           444.95    303.17    1.468 0.142639
x39          1004.17    265.12    3.788 0.000165 ***
x310          519.67    241.55    2.151 0.031787 *
x311          -116.69    230.78   -0.506 0.613257
x312           -89.59    182.21   -0.492 0.623098
x41          -589.70    180.36   -3.270 0.001130 **
x51           212.05    109.49    1.937 0.053187 .
x52           309.53    107.13    2.889 0.003982 **
x53           381.36    107.48    3.548 0.000414 ***
x54           386.34    107.53    3.593 0.000350 ***
x55           436.98    107.44    4.067 5.30e-05 ***
x56           440.46    106.56    4.133 4.01e-05 ***
x62          -462.54     77.09   -6.000 3.16e-09 ***
x63          -1965.09    197.05   -9.972 < 2e-16 ***
x7            2855.01   1398.16    2.042 0.041526 *
x8            1786.16   1462.12    1.222 0.222261
x9           -1535.47    292.45   -5.250 2.01e-07 ***
x10          -2823.30    414.55   -6.810 2.09e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 769.2 on 702 degrees of freedom
Multiple R-squared:  0.8484,    Adjusted R-squared:  0.8423
F-statistic: 140.3 on 28 and 702 DF,  p-value: < 2.2e-16
```

Figure 6: Summary of Model 1

The full multiple linear regression model was fitted using ten predictors:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_{10} x_{10} + \epsilon$$

Based on the output of the full linear model, the fitted multiple linear regression equation is:

$$\begin{aligned} \hat{y} = & 1485.84 + 884.71 * x_{12} + 832.70 * x_{13} + 1575.35 * x_{14} + 2019.74 * x_{21} + 1 \\ & 31.03 * x_{32} + 542.83 * x_{33} + 451.17 * x_{34} + 735.51 * x_{35} + 515.40 * x_{36} + 30.80 * \\ & x_{37} + 444.95 * x_{38} + 1004.17 * x_{39} + 519.67 * x_{310} - 116.69 * x_{311} - 89.59 * x_{31} \\ & 2 - 589.70 * x_{41} + 212.05 * x_{51} + 309.53 * x_{52} + 381.36 * x_{53} + 386.34 * x_{54} + 436 \\ & .98 * x_{55} + 440.46 * x_{56} - 462.54 * x_{62} - 1965.09 * x_{63} + 2855.01 * x_7 + 1786.16 \\ & * x_8 - 1535.47 * x_9 - 2823.30 * x_{10} + \epsilon \end{aligned}$$

Here:

- y = Total bike rentals
- x_{12} – x_{14} = Dummies for season
- x_{21} = Dummy for yr = 1
- x_{32} – x_{312} = Dummies for month (Feb to Dec)
- x_{41} = Dummy for holiday = 1
- x_{51} – x_{56} = Dummies for weekday (Monday to Saturday)
- x_{62} – x_{63} = Dummies for weathersit
- x_7 = Temperature
- x_8 = Feels-like temperature
- x_9 = Humidity
- x_{10} = Windspeed

Inference: The full multiple linear regression model has high explanatory power, with a R^2 of 0.8484 and an adjusted R^2 of 0.8423. The included predictor variables account for roughly 84% of the variation in daily bike rental counts. Several predictors, such as season, year, month, weekday, weather situation, temperature, and humidity, have high statistical significance (p-values less than 0.05), proving their useful contribution to the model. Notably, the factors denoting winter (x_{14}), the year 2012 (x_{21}), September (x_{39}), and severe weather conditions (x_{63}) all have a significant impact on bike demand. However, the inclusion of both temperature (x_7) and perceived temperature (x_8) raises concerns regarding potential multicollinearity, given their high correlation. The model's residual standard error is 769.2, indicating a moderate spread of residuals around expected values. The model's statistical validity is supported by an F-statistic of 140.3 and a p-value of less than $2.2e-16$.

ANOVA Table

```
> # ANOVA test
> anova(model_full)
Analysis of Variance Table

Response: y
      Df    Sum Sq   Mean Sq    F value    Pr(>F)
x1      3 950595868 316865289   535.4803 < 2.2e-16 ***
x2      1 884008263 884008263 1493.9126 < 2.2e-16 ***
x3     11 187311622 17028329   28.7767 < 2.2e-16 ***
x4      1  3306975   3306975    5.5886 0.0183495 *
x5      6 15839061  2639843    4.4612 0.0001921 ***
x6      2 185659616  92829808   156.8759 < 2.2e-16 ***
x7      1  59396300  59396300   100.3756 < 2.2e-16 ***
x8      1  3068742   3068742    5.1860 0.0230701 *
x9      1  7500820   7500820   12.6759 0.0003954 ***
x10     1  27446438  27446438   46.3826 2.092e-11 ***
Residuals 702 415401688    591740
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 7: ANOVA Table.

Inference: The Analysis of Variance (ANOVA) table shows the importance of each predictor variable in explaining the variation in bike rental counts (y). All of the predictors—x1 (season), x2 (year), x3 (month), x4 (holiday), x5 (weekday), x6 (weather situation), x7 (temperature), x8 (apparent temperature), x9 (humidity), and x10 (windspeed)—are statistically significant at various levels. Notably, x2 (year) has the highest F-statistic (1493.91) and an exceptionally modest p-value ($< 2.2e-16$), indicating a strong influence on the response variable. Variables such as x1 (season), x3 (month), and x6 (weather situation) have very high F-values (535.48, 28.78, and 156.88, respectively), indicating a considerable influence on bike rental patterns. All variables have p-values < 0.05 , showing statistical significance. However, x4 (holiday) and x8 (apparent temperature) have smaller F-values (5.59 and 5.19), implying they provide less explanatory power than others. Overall, the ANOVA reveals that each predictor significantly contributes to the model, with year, season, and weather-related factors having the greatest influence on explaining variations in bike rental demand.

Confidence Interval of coefficients

```
> confint(model_full)
              2.5 %      97.5 %
(Intercept) 1015.137895 1956.5500
x12          532.304504 1237.1171
x13          414.254762 1251.1497
x14         1219.982881 1930.7184
x21         1905.428528 2134.0423
x32         -151.257695  413.3079
x33          218.025196  867.6303
x34          -34.894526  937.2258
x35          210.058893 1260.9521
x36          -39.065944 1069.8755
x37         -585.343482  646.9368
x38         -150.270031 1040.1679
x39          483.644265 1524.7014
x310          45.420268  993.9283
x311         -569.787515  336.4010
x312         -447.342146  268.1591
x41          -943.811261 -235.5831
x51          -2.921253  427.0290
x52          99.186216  519.8691
x53         170.341109  592.3737
x54         175.226839  597.4591
x55         226.042864  647.9247
x56         231.239788  649.6779
x62         -613.886658 -311.1896
x63        -2351.967857 -1578.2052
x7           109.942053  5600.0794
x8        -1084.493648  4656.8084
x9        -2109.646448 -961.2904
x10        -3637.207096 -2009.3863
```

Figure 8: Confidence Intervals.

Inference: The 95% confidence intervals for the regression coefficients in the full model provide useful information about the reliability and relevance of each predictor's impact on daily bike rental numbers. Several variables, including seasonal indicators (x12, x13, x14), the year (x21), specific months such as March, May, September, and October (x33, x35, x39, x310), and weekdays from Tuesday to Saturday (x52-x56), have confidence intervals that do not include zero, confirming their statistically significant and positive effect on rental numbers. Notably, the year 2012 (x21) has a continuously large positive influence, whereas weather condition indicators such as x62 and x63 have considerable negative effects, showing that terrible weather severely affects demand.

Environmental variables such as temperature (x7) have an obvious positive impact, whilst humidity (x9) and windspeed (x10) have considerable negative effects on rental counts. The variable x41 (holiday) likewise exhibits a negative and statistically significant effect, indicating that fewer bikes are hired on holidays. On the other hand, several month-related dummy variables—February, April, June, July, August, November, and December—as well as Monday (x51) and apparent temperature (x8) have wide confidence intervals that include zero, indicating they are not statistically significant contributors in the presence of other predictors. The wide and overlapping confidence intervals for both temperature and perceived temperature indicate potential multicollinearity, which may impair the reliability of their separate effects. Overall, the confidence intervals indicate that season, year, temperature, weather conditions, significant months, and weekdays are all strong predictors of bike rental demand.

Residual Analysis

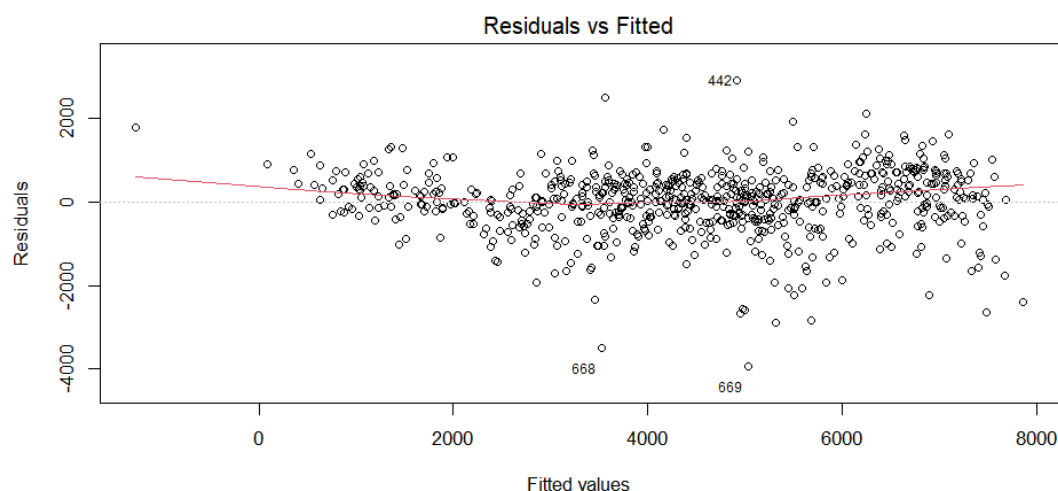


Figure 9: Residuals vs Fitted Plot of the Residual

Inference: The residuals are distributed randomly about the horizontal line at zero, showing that the linearity condition is reasonably met. However, the spread of residuals increases slightly at higher fitted values, indicating minor heteroscedasticity, which occurs when the variance of the errors does not remain constant across all levels of the response. Furthermore, a few key observations, such as points 442, 668, and 669, stand out as outliers and may require further investigation. Overall, the graph supports the linear model's adequacy while also highlighting regions where model improvement or further diagnostics could be advantageous.

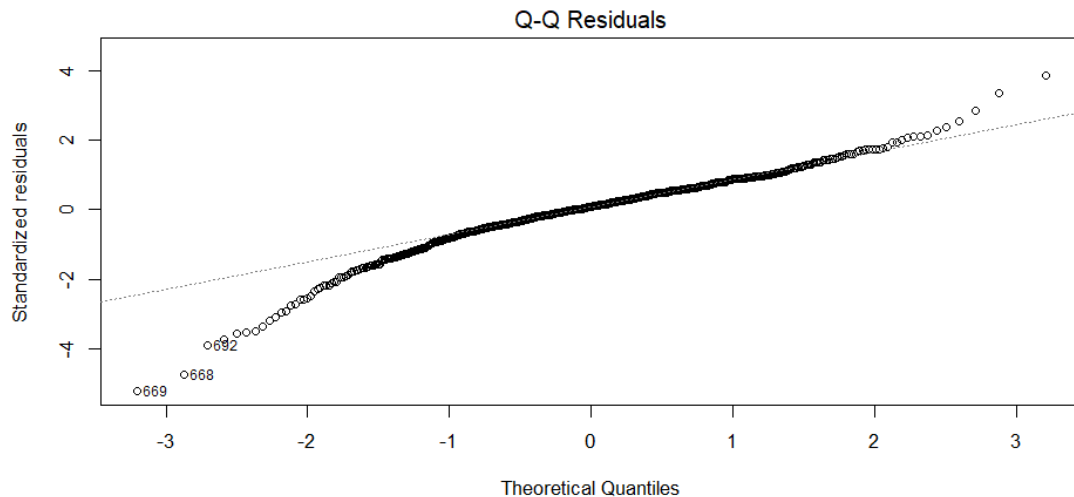


Figure 10: Q-Q Plot of the Residuals.

Inference: Most of the points in this plot are roughly aligned with the QQ line, indicating that the residuals are approximately normally distributed. However, there are significant deviations at both ends of the plot, particularly in the lower tail, where a few residuals, such as points 668, 669, and 692, are substantially below the line. These indicate the presence of mild non-normality or larger tails in the distribution. Despite these outliers, the overall shape of the plot confirms the assumption of normality for most data points, and the model is still statistically valid, especially given the huge sample size.

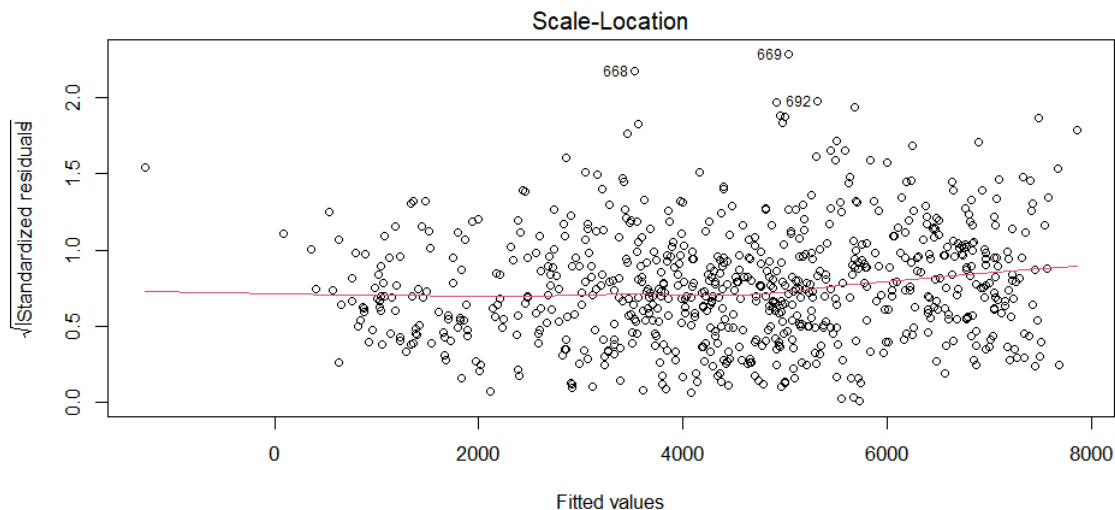


Figure 11: Scale-Location plot of the Residual.

Inference: In this plot, the residuals appear to be rather uniformly distributed across the range of fitted values, with no apparent funnel shape or systematic trend. This implies that the premise of constant variance is reasonably satisfied. However, there is a tiny rising trend and increased dispersion as fitted values increase, showing minor heteroscedasticity. Several observations, including 668, 669, and 692, have relatively high standardized residuals, which could indicate probable outliers or influencing observations. Overall, the plot confirms the assumption of homoscedasticity, except for a slight variance increase at higher fitted values.

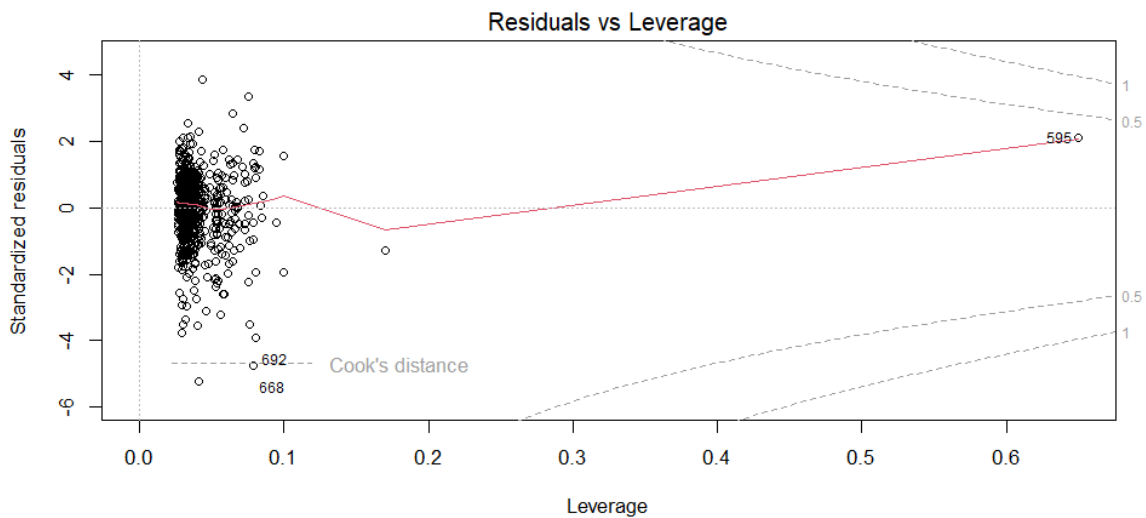


Figure 12: Residuals vs Leverage plot

Inference: The majority of the data in this graph are clustered within low-leverage and moderate-residual categories, indicating that they do not have an undue influence on the model. However, observation 595 stands out with high leverage and moderate residuals, placing it toward the outer Cook's distance curve, implying that it may have a significant impact on the fitted model. Additionally, points 668 and 692 have significant residuals but lower leverage, indicating that they are outliers but not necessarily influential. Overall, the model does not appear to be too influenced by extreme leverage points, but a few findings may require further investigation to maintain model robustness.

Statistical Tests

Non-Constant Variance (NCV) Test

H_0 : The variance of residuals is constant (homoscedasticity is present).

H_1 : The variance of residuals is not constant (heteroscedasticity is present).

```
> ncvTest(model_full)
Non-constant Variance Score Test
Variance formula: ~ fitted.values
Chisquare = 16.59102, Df = 1, p = 4.637e-05
```

Figure 13: NCV test output.

Inference: The test results indicate a Chi-square statistic of 16.59102, with one degree of freedom and a p-value of 4.637e-05. Because the p-value is much lower than the standard alpha level of 0.05, we reject the null hypothesis. This result gives strong evidence for heteroscedasticity in the model, which means that the variance of the residuals varies with the level of the fitted values. This violates one of linear regression's major assumptions, potentially affecting the dependability of standard errors and p-values.

Normality Test

H_0 : The residuals are normally distributed.

H_1 : The residuals are not normally distributed.

```

Shapiro-Wilk normality test

data:  model_full$residuals
W = 0.9518, p-value = 1.011e-14

```

Figure 14: Shapiro-Wilk Test output.

Inference: The test yielded a W statistic of 0.9518 and a p value of 1.011e-14. Since the p-value is significantly lower than the threshold of 0.05, we reject the null hypothesis that the residuals are normally distributed. This finding violates the normality assumption.

Autocorrelation Test

H₀: There is no autocorrelation in the residuals ($\rho = 0$).

H₁: There is autocorrelation in the residuals ($\rho \neq 0$).

```

> durbinWatsonTest(model_full)
lag Autocorrelation D-W Statistic p-value
1      0.4006853      1.198164      0
Alternative hypothesis: rho ≠ 0

```

Figure 15: DurbinWatson Test output.

Inference: We reject the null hypothesis since the p-value is 0 and is significantly lower than the 0.05 threshold. This shows that the residuals show positive autocorrelation. The Durbin-Watson statistic of 1.1982, which is significantly lower than the optimal value of 2, lends support to this conclusion. The presence of autocorrelation contradicts a major premise of linear regression and may result in skewed standard errors, rendering inference inaccurate. This suggests that time series structure or lagged factors should be included in the model.

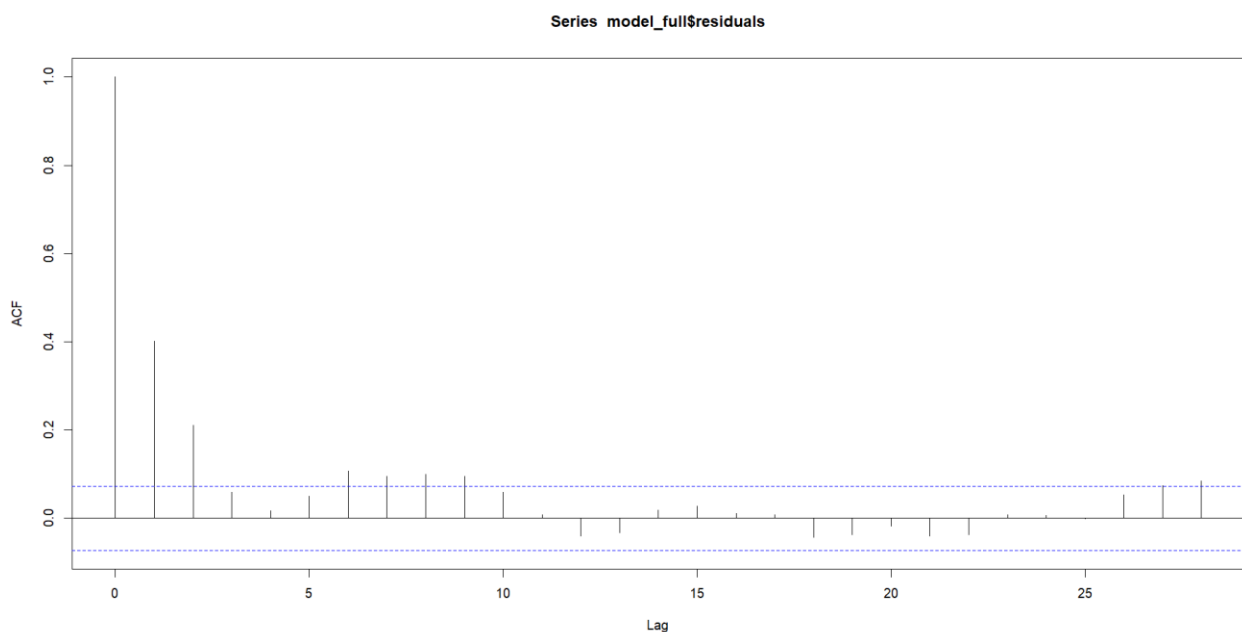


Figure 16: ACF Plot of the Residuals.

Inference: The Autocorrelation Function (ACF) plot of the residuals from the full regression model reveals considerable autocorrelation at lag 1 and many subsequent lags. The bars for these lags clearly surpass the blue significance thresholds, demonstrating that the residuals do not vary randomly over time. This pattern is a strong indicator of autocorrelation, implying that past errors are associated with future ones, which violates linear regression's independence premise. This verifies the Durbin-Watson test result and emphasizes the need to add time series modelling techniques.

Multi-collinearity Test

```
> vif(model_full)
      GVIF Df GVIF^(1/(2*Df))
x1  169.905085 3      2.353429
x2   1.046828 1      1.023146
x3  408.760454 11     1.314326
x4   1.121296 1      1.058913
x5   1.163210 6      1.012678
x6   1.894831 2      1.173256
x7  80.806689 1      8.989254
x8  70.036722 1      8.368795
x9   2.140362 1      1.462998
x10  1.273296 1      1.128404
```

Figure 17: VIF Score.

Inference: The VIF analysis for the entire model shows that most predictor variables have acceptable multicollinearity levels, with GVIF-adjusted values less than the standard threshold of 5. However, two variables—x7 (temperature) and x8 (apparent temperature)—stand out for their significantly high VIF values of 8.99 and 8.37, respectively. These high scores imply considerable multicollinearity between the two, most likely because they are closely related measurements of environmental temperature. Multicollinearity can distort coefficient estimates and diminish the reliability of inferences. As a result, removing one of these variables or experimenting with dimension reduction approaches may improve model interpretability and stability.

Model Adequacy (Model 1)

Strengths:

- The model has a high R^2 (0.8484) and modified R^2 (0.8423), showing that it accounts for a significant percentage of the variance in bike rental counts.
- Most predictors are statistically significant, with year, season, and weather-related variables contributing significantly to the model.
- Key variables (e.g., year 2012, winter, temperature, humidity) have significant confidence intervals that do not include zero, indicating their importance.

Concerns:

- Heteroscedasticity: The NCV test ($p = 4.637e-05$) and residual plots show non-constant variance, which contradicts the premise of homoscedasticity and impairs the reliability of standard errors and confidence intervals.
- Non-normality of residuals: The Shapiro-Wilk test ($p < 0.05$) and Q-Q plot demonstrate deviations from normality, especially in the tails, potentially compromising the validity of hypothesis testing for coefficients.

- Autocorrelation: Strong positive autocorrelation is revealed by the Durbin-Watson statistic (1.1982) and the ACF plot, which defies the assumption of independent residuals and shows that time-related structure in the data is not taken into consideration.
- Multicollinearity: High apparent temperature (x8) and temperature (x7) VIF values indicate significant multicollinearity, which could skew standard errors and make the model harder to interpret.

While the model captures a great amount of data variance and many predictors are statistically significant, its violations of important assumptions—particularly independence, homoscedasticity, and multicollinearity—limit its potential as a reliable model. To increase the model's resilience and inference quality, these difficulties must be addressed by variable selection, time series modelling (for example, by adding lag or trend components), and the removal or transformation of associated predictors.

MODEL 2 (Subset Selection)

To enhance the initial model and reduce complexity, all subsets regression was conducted using the `regsubsets()` function. This function evaluates predictor combinations to maximize adjusted R^2 , a statistic that balances model fit with parsimony.

```
> summary_r
Subset selection object
Call: regsubsets.formula(y ~ x1 + x2 + x3 + x4 + x5 + x6 + x7 + x9 +
  x10, data = bike_clean, nvmax = 10)
27 Variables (and intercept)
      Forced in Forced out
x12      FALSE      FALSE
x13      FALSE      FALSE
x14      FALSE      FALSE
x21      FALSE      FALSE
x32      FALSE      FALSE
x33      FALSE      FALSE
x34      FALSE      FALSE
x35      FALSE      FALSE
x36      FALSE      FALSE
x37      FALSE      FALSE
x38      FALSE      FALSE
x39      FALSE      FALSE
x310     FALSE      FALSE
x311     FALSE      FALSE
x312     FALSE      FALSE
x41      FALSE      FALSE
x51      FALSE      FALSE
x52      FALSE      FALSE
x53      FALSE      FALSE
x54      FALSE      FALSE
x55      FALSE      FALSE
x56      FALSE      FALSE
x62      FALSE      FALSE
x63      FALSE      FALSE
x7       FALSE      FALSE
x9       FALSE      FALSE
x10      FALSE      FALSE
1 subsets of each size up to 10
```

Figure 18: Summary of regression subsets

The Adjusted R² method selects a linear regression model with reduced significant variables to maintain model performance and improve parsimony.

The fitted regression equation is:

$$\hat{y} = 1713.39 + 779.21 \cdot x_{1_2} + 1252.55 \cdot x_{1_4} + 1982.24 \cdot x_{2_1} + 891.80 \cdot x_{3_9} - 750.90 \cdot x_{4_1} - 391.99 \cdot x_{6_2} - 1810.84 \cdot x_{6_3} + 6249.27 \cdot x_7 - 1747.14 \cdot x_9 - 2985.09 \cdot x_{10} + \epsilon$$

Here:

- y = Total bike rentals
- x1_2, x1_4 = Dummy variables for season (summer and winter)
- x2_1 = Dummy for year 2012
- x3_9 = Dummy for month September
- x4_1 = Dummy for holiday
- x6_2, x6_3 = Dummy variables for weather conditions
- x7 = Temperature
- x9 = Humidity
- x10 = Windspeed

Inference: The regression model chosen using best subset regression has excellent predictive performance. With a Multiple R² of 0.8285 and an Adjusted R² of 0.8261, this model explains approximately 82.6% of the variation in daily bike rentals using only ten selected predictors. All variables in the model are statistically significant (p < 0.001), indicating their important contribution to the model. Positive coefficients for x12 (summer), x14 (winter), x21 (year 2012), x39 (September), and x7 (temperature) indicate that these variables are associated to increased bike demand. Conversely, negative coefficients for x41 (holiday), x62 & x63 (poor weather conditions), x9 (humidity), and x10 (windspeed) show that bad weather and holidays diminish bike usage. The residual standard error of 807.8 is fair, and the F-statistic of 347.8 with a p-value < 2.2e-16 validates the overall model's strong statistical significance. In conclusion, the reduced model retains great explanatory power with fewer variables, enhancing interpretability while keeping strong predictive ability.

ANOVA Table

```
> anova(model_Rsquare)
Analysis of Variance Table

Response: y
      Df    Sum Sq   Mean Sq    F value    Pr(>F)
x1_2    1  58554017  58554017    89.732 < 2.2e-16 ***
x1_4    1  39715401  39715401    60.862 2.154e-14 ***
x2_1    1  880601480 880601480 1349.487 < 2.2e-16 ***
x3_9    1 144382197 144382197  221.260 < 2.2e-16 ***
x4_1    1  14008849  14008849   21.468 4.270e-06 ***
x6_2    1 104611054 104611054  160.312 < 2.2e-16 ***
x6_3    1 159230787 159230787  244.015 < 2.2e-16 ***
x7      1 824221617 824221617 1263.087 < 2.2e-16 ***
x9      1  11278347  11278347   17.284 3.607e-05 ***
x10     1  33099060  33099060   50.723 2.583e-12 ***
Residuals 720 469832584    652545
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 22: ANOVA Table of Model 2.

Inference: The ANOVA analysis for this regression model shows that all specified factors significantly explain fluctuations in daily bike rental numbers. Each variable has a p-value of less than 0.001, indicating great statistical relevance. Among them, the year 2012 (x2_1) has the greatest F-value (1349.49), making it the most influential component in the model. Temperature (x7) and severe weather conditions (x6_3) also have unusually high F-values, indicating a significant impact on bike demand. Other predictors, including summer (x1_2), September (x3_9), and winter (x1_4), support the importance of seasonal and temporal dynamics. Although variables like holiday (x4_1) and humidity (x9) have lower F-values, they nonetheless make a significant contribution to the model. Overall, the ANOVA results confirm the power and relevance of the chosen predictors, underlining the significance of time-related and contextual factors in predicting bike rental behaviour.

Confidence Intervals of the Coefficients

```
> confint(model_Rsquare)
```

	2.5 %	97.5 %
(Intercept)	1286.0850	2140.7019
x1_2	631.4552	926.9702
x1_4	1100.6351	1404.4587
x2_1	1863.4771	2101.0047
x3_9	665.6649	1117.9418
x4_1	-1102.6669	-399.1421
x6_2	-548.1574	-235.6371
x6_3	-2209.1703	-1412.5123
x7	5892.2132	6606.3283
x9	-2319.5097	-1174.7614
x10	-3807.9602	-2162.2129

Figure 23: Confidence Intervals of the coefficients of Model 2.

Inference: The 95% confidence intervals for the coefficients in the simplified regression model show that all of the specified factors have a statistically significant effect on bike rental demand. Notably, none of the intervals include zero, indicating that each variable makes a meaningful contribution to the response variable. For example, the year indicator x2_1 (2012) has a narrow and completely positive interval (1863.48 to 2101.00), indicating a significant and constant growth in rentals that year. Similarly, environmental variables such as temperature (x7) have a big positive influence, whereas humidity (x9) and windspeed (x10) have huge negative intervals, showing that severe weather conditions considerably reduce bike rentals. Categorical variables with tight, significant intervals include season (x1_2, x1_4), September (x3_9), and severe weather (x6_2, x6_3). Overall, the confidence intervals highlight the dependability and significance of each predictor in this model.

Residual Analysis

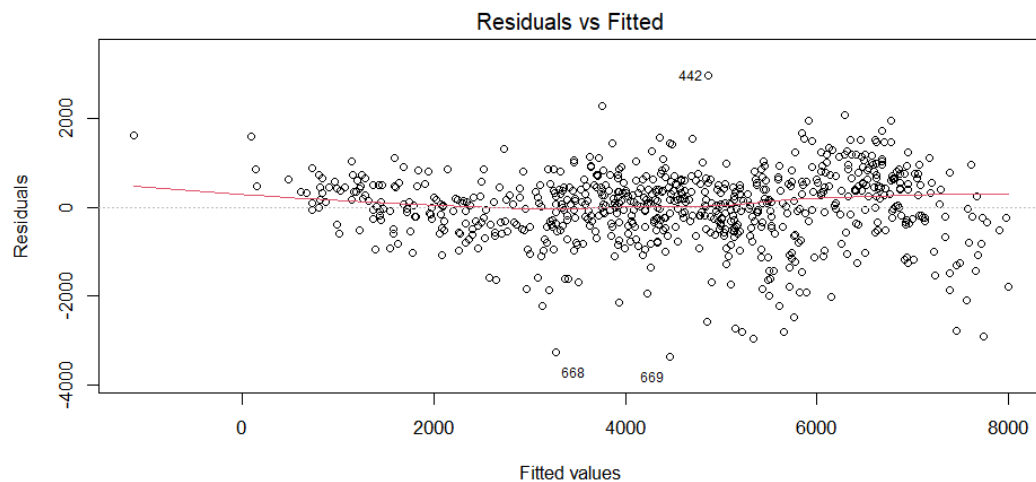


Figure 24: Residual Vs Fitted Plot.

Inference: The "Residuals vs Fitted" plot for the reduced model reveals that residuals are randomly distributed about the horizontal line at zero, implying that the linearity requirement is reasonably met. However, the red smoothed line has a slight upward curve, which could indicate non-linearity that the model could not capture. There is also a minor funnel-shaped pattern, which suggests possible heteroscedasticity—the variance of residuals may grow with fitted values. A few outliers, including observations 442, 668, and 669, depart significantly from the main cluster and may be influential. Overall, while the model performs well, slight breaches of assumptions such as homoscedasticity and linearity may persist.

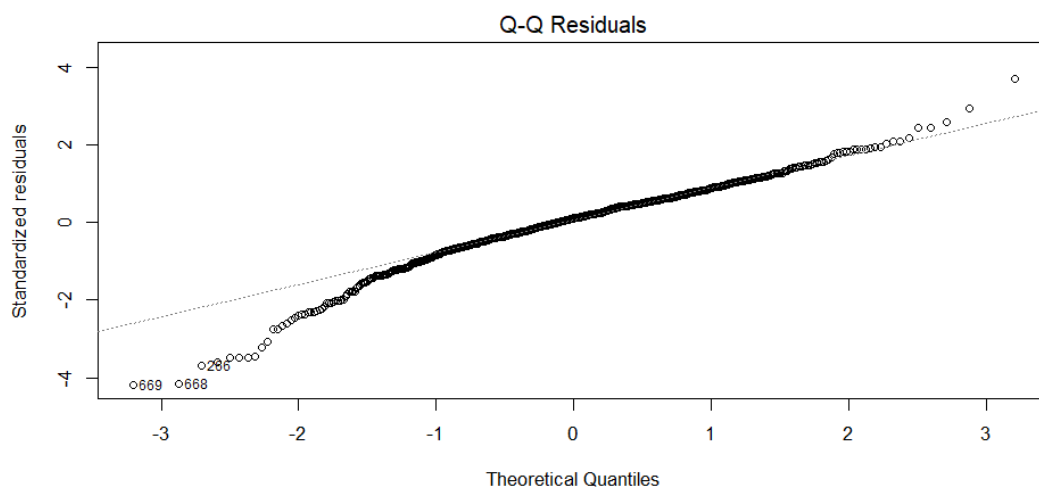


Figure 25: QQ Plot of the Residual.

Inference: The Q-Q plot of residuals for the reduced model indicates that the residuals are roughly normally distributed, as the majority of dots fall along the 45-degree reference line. However, there are small variations in both tails, notably in the lower tail, showing minor departures from normality. Observations 668 and 669 deviate significantly from the line, indicating the existence of outliers or heavier tails than predicted for a normal distribution. Despite these modest deviations, the overall pattern supports the assumption of normality for most of the data, and the model is still viable for inference in large-sample settings.

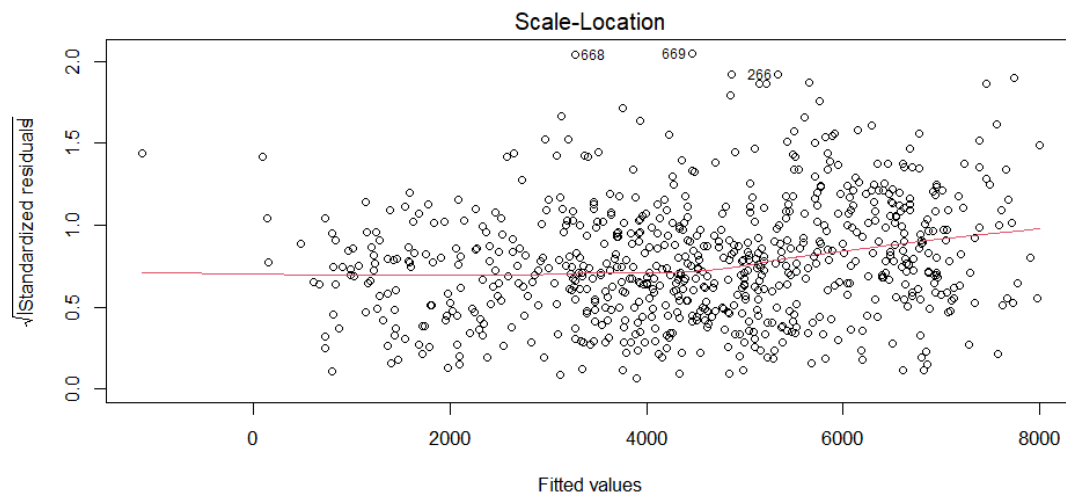


Figure 26: Scale-Location Plot of the Residuals

Inference: The Scale-Location plot is used to evaluate the homoscedasticity assumption in linear regression, which states that the variance of the residuals should be constant across all levels of fitted values. The standardized square root of the residuals appears to be fairly evenly spread across the range of fitted values, with no discernible strong curvature or funnelling; however, the red smoothing line shows a slight upward trend, particularly at higher fitted values, which may indicate minor heteroscedasticity; some observations, such as points 668, 669, and 266, have higher standardized residuals, which may indicate outliers or significant cases. Overall, the plot supports the assumption of constant variance, with the exception of a little suggestion that variability may somewhat rise with higher fitted values, implying that further checks or robust regression approaches should be considered to improve model dependability.

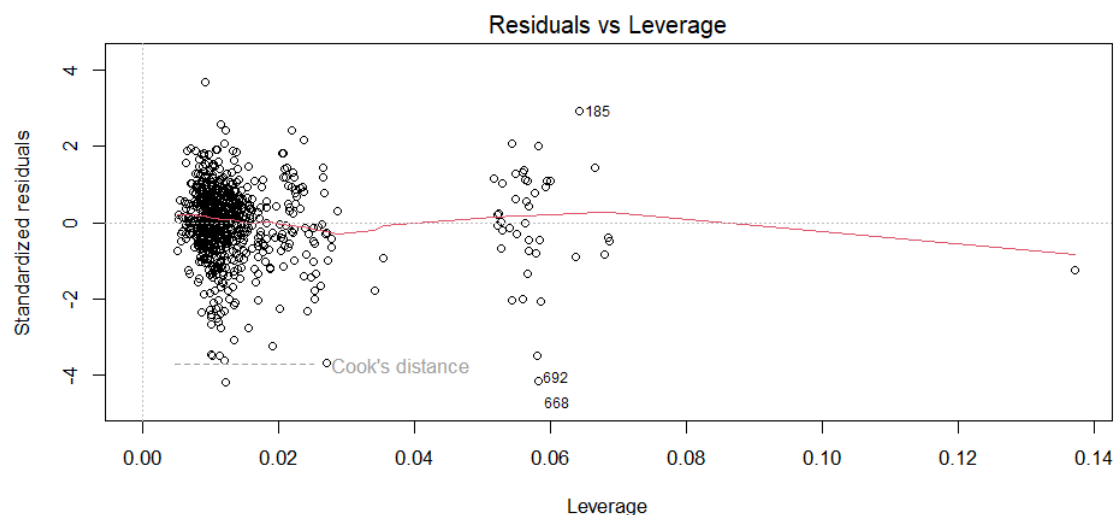


Figure 27: Residual Vs Leverage Plot.

Inference: In this plot, the majority of data points cluster around low leverage and moderate standardized residuals, indicating that they have little influence on the model. However, a few observations stand out, including points 185, 668, and 692. Point 185 has a high leverage and a moderate residual, putting it on the outer Cook's distance curve, implying that it could have a major impact on the regression coefficients. Points 668 and 692 have large standardized residuals, indicating

potential outliers, although being within reasonable leverage ranges. Overall, while the bulk of data points do not appear to be problematic, the highlighted observations require further investigation to determine whether they have a disproportionate effect on the model's accuracy or stability.

Statistical Tests

Non-Constant Variance (NCV) Test

H₀: The variance of residuals is constant (homoscedasticity is present).

H₁: The variance of residuals is not constant (heteroscedasticity is present).

```
> ncvTest(model_Rsquare)
Non-constant Variance Score Test
Variance formula: ~ fitted.values
Chisquare = 32.65874, Df = 1, p = 1.0984e-08
```

Figure 28: NCV Test Output.

Inference: The NCV test determines if the variance of residuals is consistent (homoscedastic) across all fitted values. The test results show a Chi-square value of 32.65874 with one degree of freedom and a p-value of 1.0984e-8. Because the p-value is significantly lower than 0.05, we reject the null hypothesis. This suggests substantial evidence of heteroscedasticity in the model's residuals, which contradicts a crucial linear regression assumption. This may alter the validity of standard errors and confidence intervals, implying that corrective methods (e.g., transformation or weighted least squares) may be required.

Normality Test

H₀: The residuals are normally distributed.

H₁: The residuals are not normally distributed.

```
Shapiro-Wilk normality test

data:  model_Rsquare$residuals
W = 0.96463, p-value = 2.74e-12
```

Figure 29: Shapiro-Wilk Test output.

Inference: The Shapiro-Wilk normality test was used to determine whether the residuals from the selected regression model have a normal distribution. The test yielded a W statistic of 0.96463 and a p-value of 2.74 e-12. We reject the null hypothesis of normally distributed residuals because the p-value is much lower than the usual threshold of 0.05. This finding indicates a violation of the normality assumption in linear regression, implying the presence of non-normal residuals. While linear regression can be somewhat resistant to this assumption in big datasets, deviations from normality can still influence the quality of statistical inference, such as confidence intervals and significance tests.

Autocorrelation Test

H₀: There is no autocorrelation in the residuals ($\rho = 0$).

H₁: There is autocorrelation in the residuals ($\rho \neq 0$).

```
> durbinWatsonTest(model_Rsquare)
lag Autocorrelation D-W Statistic p-value
1 0.4326262 1.133545 0
Alternative hypothesis: rho ≠ 0
```

Figure 30: Durbin Watson Test output.

Inference: The Durbin-Watson test was used to determine whether or not the regression model's residuals had autocorrelation. The test produced a Durbin-Watson statistic of 1.1335 and a p-value of zero. We reject the null hypothesis of no autocorrelation because the p-value is significantly less than 0.05. The statistic is significantly lower than 2, indicating positive autocorrelation, implying that the residuals are not independent and are likely influenced by their preceding values. This contradicts a basic premise of linear regression, implying that time-related or lagged variables may be required to increase model dependability.

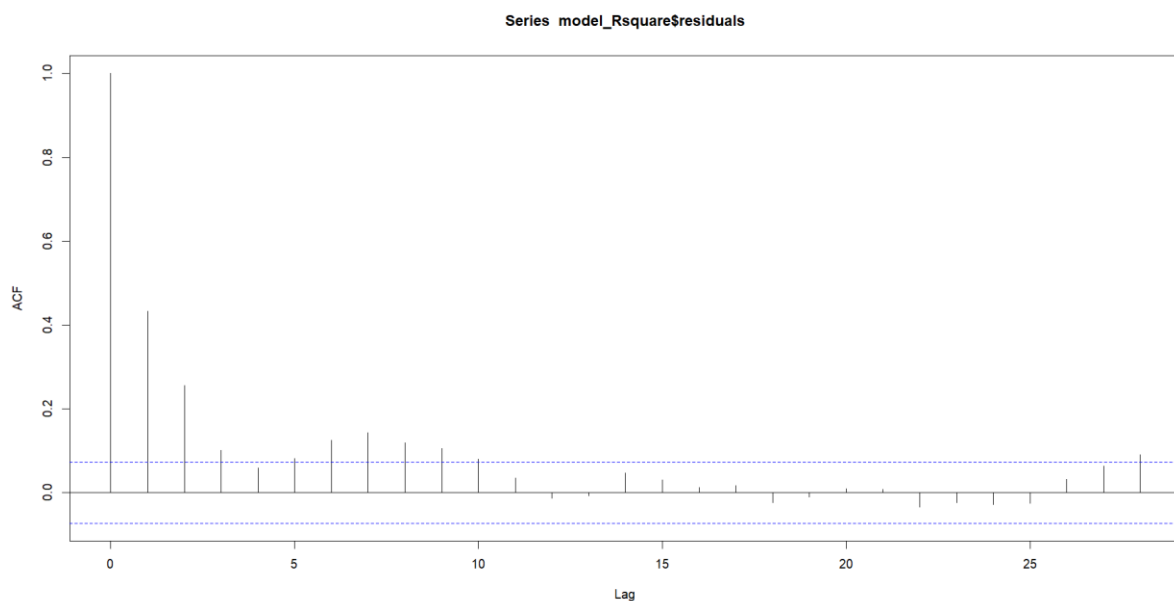


Figure 31: ACF Plot of Residual

Inference: The ACF (Autocorrelation Function) plot of the residuals from the model_Rsquare regression model shows a considerable autocorrelation at lag 1, which steadily decreases at later lags. Several of these bars exceed the significance levels (shown by the blue dotted lines), particularly at lower lags, implying that residuals do not vary randomly over time. This violates the independence condition in regression modelling and validates the Durbin-Watson test results. The presence of autocorrelation shows that a time series structure exists in the data, and integrating lagged elements or employing time series modelling techniques may increase model performance and inference accuracy.

Multi-collinearity Test

```
> vif(model_Rsquare)
x1_2 x1_4 x2_1 x3_9 x4_1 x6_2 x6_3 x7 x9 x10
1.195140 1.235500 1.024841 1.119791 1.003439 1.587645 1.286696 1.239868 1.928911 1.180322
```

Figure 32: VIF Score.

Inference: The VIF (Variance Inflation Factor) values for all predictors in the model_Rsquare are far lower than the widely accepted threshold of 5, indicating that

multicollinearity is not an issue in this model. Specifically, the VIF scores vary from about 1.00 to 1.93, indicating that none of the independent variables are substantially linearly associated with one another. This improves the reliability of coefficient estimations while also contributing to the regression model's stability and interpretability.

Model Adequacy (Model 2)

Strengths:

- **Efficient Model Performance:** Model 2 has a high R^2 of 0.8285 and an adjusted R^2 of 0.8261, indicating excellent explanatory power with fewer predictors than the entire model.
- **All predictors are statistically significant.** The included variables have p-values < 0.001 , indicating their significance. These include seasonal indications (x1_2, x1_4), the year 2012 (x2_1), September (x3_9), meteorological conditions (x6_2, x6_3), and continuous variables like temperature (x7), humidity (x9), and windspeed (x10).
- **Reliable coefficient estimates:** The confidence intervals for all coefficients exclude 0, confirming their strong impact on bike rental counts.
- **No multicollinearity:** The VIF values for all predictors are significantly lower than the conventional threshold of 5, indicating that multicollinearity is not a concern in this model.

Concerns:

- **Heteroscedasticity persists:** The NCV test ($p = 1.0984e-08$) and the scale-location plot show non-constant residual variance, implying heteroscedasticity and possible inefficiency in standard error estimates.
- **Non-normal residuals:** The Shapiro-Wilk test ($p = 2.74e-12$) and the Q-Q plot reveal deviations from normality in the residual distribution, especially in the tails.
- **Autocorrelation remains an issue.** The Durbin-Watson statistic (1.1335), low p-value, and ACF plot all corroborate the presence of positive autocorrelation, meaning that residuals are not independent and that the data's time structure is not completely preserved.
- **Slight heteroscedasticity and outliers:** Plots such as residuals versus fitted, scale-location, and residuals vs leverage reveal modest breaches of assumptions as well as the presence of influential observations (e.g., points 668, 669, 185), which may have an impact on model stability.

Model 2 is simpler and more interpretable than the entire model, while still retaining high predictive capability. It successfully reduces multicollinearity, leaving only the most important predictors. However, assumptions about homoscedasticity, normality, and residual independence are still violated, indicating that the model must be refined further.

MODEL 3 (GLS Model)

To address the occurrence of autocorrelation in the residuals found in earlier linear models, a Generalized Least Squares (GLS) model was fitted with an AR(1) correlation structure. A time index variable was included to reflect the temporal character of the data. The model has ten predictors (x1–x10) that represent seasonal, yearly, weather, and temperature-related factors.

Model Summary

```
> summary(model_gls)
Generalized least squares fit by REML
Model: y ~ x1 + x2 + x3 + x4 + x5 + x6 + x7 + x8 + x9 + x10
Data: bike_clean
      AIC      BIC    logLik
11340.46 11481.63 -5639.229

Correlation Structure: AR(1)
Formula: ~time_index
Parameter estimate(s):
      Phi
0.4562165
```

Figure 33: GLS Model Summary.

Coefficients:				
	Value	Std.Error	t-value	p-value
(Intercept)	1498.661	264.8057	5.659475	0.0000
x12	864.392	260.4226	3.319190	0.0009
x13	823.007	307.2536	2.678593	0.0076
x14	1304.567	262.0801	4.977742	0.0000
x21	1989.986	95.6078	20.814056	0.0000
x32	69.162	222.7606	0.310477	0.7563
x33	530.484	250.3685	2.118812	0.0345
x34	369.532	351.6199	1.050941	0.2936
x35	620.871	374.8960	1.656115	0.0981
x36	314.333	383.2332	0.820214	0.4124
x37	-154.394	426.3395	-0.362138	0.7174
x38	309.992	416.9199	0.743529	0.4574
x39	941.851	375.0001	2.511602	0.0122
x310	685.553	350.3507	1.956763	0.0508
x311	155.972	340.2843	0.458358	0.6468
x312	135.143	274.9372	0.491540	0.6232
x41	-390.375	147.6623	-2.643699	0.0084
x51	173.825	83.1007	2.091743	0.0368
x52	290.480	97.0272	2.993798	0.0029
x53	363.138	102.8016	3.532416	0.0004
x54	351.837	102.8754	3.420032	0.0007
x55	414.244	97.1515	4.263900	0.0000
x56	421.289	80.4836	5.234472	0.0000
x62	-406.768	66.6358	-6.104354	0.0000
x63	-1795.644	173.3353	-10.359370	0.0000
x7	3166.230	1285.3783	2.463267	0.0140
x8	2032.975	1299.4361	1.564506	0.1181
x9	-1989.174	278.7674	-7.135605	0.0000
x10	-2317.833	378.7230	-6.120128	0.0000

Figure 34: GLS Model Summary.

Correlation:															
	(Intr)	x12	x13	x14	x21	x32	x33	x34	x35	x36	x37	x38	x39	x310	x311
x12	0.034														
x13	0.041	0.573													
x14	-0.014	0.316	0.553												
x21	-0.207	-0.005	0.003	-0.004											
x32	-0.352	-0.010	0.014	0.018	0.016										
x33	-0.299	-0.369	-0.175	-0.087	0.035	0.484									
x34	-0.197	-0.707	-0.371	-0.199	0.038	0.349	0.641								
x35	-0.127	-0.672	-0.343	-0.180	0.047	0.346	0.634	0.806							
x36	-0.131	-0.585	-0.462	-0.247	0.060	0.347	0.616	0.755	0.803						
x37	-0.109	-0.402	-0.636	-0.343	0.058	0.312	0.513	0.599	0.649	0.777					
x38	-0.112	-0.402	-0.662	-0.363	0.052	0.311	0.505	0.587	0.633	0.756	0.866				
x39	-0.118	-0.378	-0.629	-0.478	0.044	0.331	0.506	0.572	0.611	0.722	0.820	0.830			
x310	-0.142	-0.227	-0.380	-0.703	0.033	0.330	0.432	0.439	0.460	0.529	0.581	0.586	0.681		
x311	-0.196	-0.230	-0.389	-0.731	0.023	0.321	0.403	0.405	0.406	0.463	0.514	0.524	0.622	0.790	
x312	-0.261	-0.178	-0.307	-0.580	0.023	0.383	0.438	0.403	0.400	0.442	0.475	0.486	0.574	0.708	0.722
x41	-0.040	0.009	0.011	0.032	0.006	0.008	0.024	0.008	0.010	0.027	0.016	0.027	0.003	-0.011	-0.031
x51	-0.114	-0.035	-0.023	-0.006	0.004	0.007	0.025	0.042	0.048	0.048	0.053	0.035	0.048	0.016	0.017
x52	-0.148	-0.010	-0.028	-0.011	0.010	0.001	0.010	0.034	0.033	0.054	0.069	0.055	0.061	0.032	0.009
x53	-0.160	-0.030	-0.022	0.007	0.010	-0.019	0.009	0.040	0.047	0.050	0.066	0.042	0.049	0.013	-0.009
x54	-0.192	-0.024	-0.045	-0.008	0.019	-0.006	0.002	0.042	0.044	0.059	0.087	0.066	0.058	0.029	0.002
x55	-0.203	-0.002	0.001	0.018	0.018	-0.011	-0.010	0.007	0.021	0.019	0.031	0.017	0.012	-0.002	-0.025
x56	-0.170	-0.017	-0.030	0.016	0.014	-0.012	-0.013	0.007	0.011	0.006	0.025	0.018	0.008	-0.018	-0.022
x62	0.228	-0.037	-0.020	-0.005	-0.043	-0.011	-0.012	-0.008	0.007	-0.029	-0.029	-0.020	-0.008	-0.016	0.017
x63	0.213	-0.030	-0.031	-0.017	-0.002	-0.009	-0.029	-0.002	0.008	-0.018	-0.026	-0.008	-0.011	-0.024	-0.003
x7	0.054	0.000	-0.048	-0.017	-0.049	-0.025	-0.093	-0.097	-0.183	-0.230	-0.221	-0.229	-0.182	-0.101	-0.039
x8	-0.161	-0.010	0.005	-0.008	0.013	-0.026	-0.008	-0.011	0.030	0.049	0.044	0.072	0.043	0.002	-0.010
x9	-0.553	-0.035	-0.004	0.004	0.088	0.019	0.063	0.079	0.037	0.102	0.089	0.058	0.009	-0.006	-0.006
x10	-0.390	0.016	0.040	0.089	0.028	0.003	0.004	-0.009	0.054	0.063	0.072	0.065	0.045	-0.009	-0.032

Figure 35: GLS Model Summary.

	x312	x41	x51	x52	x53	x54	x55	x56	x62	x63	x7	x8	x9
x12													
x13													
x14													
x21													
x32													
x33													
x34													
x35													
x36													
x37													
x38													
x39													
x310													
x311													
x312													
x41	-0.014												
x51	0.012	-0.256											
x52	0.016	-0.016	0.586										
x53	0.001	-0.017	0.452	0.676									
x54	-0.001	-0.025	0.391	0.533	0.689								
x55	-0.020	-0.022	0.343	0.444	0.527	0.674							
x56	-0.032	-0.001	0.273	0.350	0.397	0.463	0.599						
x62	0.000	0.006	-0.051	-0.068	-0.045	-0.080	-0.097	-0.066					
x63	-0.005	0.046	-0.044	-0.066	-0.095	-0.070	-0.034	-0.076	0.401				
x7	-0.027	-0.060	-0.004	-0.040	-0.064	-0.047	-0.082	-0.020	0.037	0.013			
x8	-0.011	0.052	-0.019	0.007	0.034	0.011	0.064	0.017	0.016	0.029	-0.929		
x9	-0.028	0.043	0.006	0.028	0.019	0.091	0.078	0.062	-0.554	-0.403	-0.026	-0.047	
x10	-0.009	0.035	-0.010	0.004	0.030	0.018	0.041	-0.010	-0.085	-0.158	-0.260	0.240	0.174
Standardized residuals:													
	Min	Q1		Med		Q3		Max					
	-4.78699566	-0.43733907		0.09204719		0.60050532		3.69379510					
Residual standard error: 787.187													
Degrees of freedom: 731 total; 702 residual													
>													

Figure 36: GLS Model Summary.

The fitted equation is:

$$\begin{aligned} \text{ybar} = & 1498.66 + 864.39 \cdot x_{12} + 823.01 \cdot x_{13} + 1304.57 \cdot x_{14} + 1989.99 \cdot x_{21} \\ & + 530.48 \cdot x_{33} + 941.85 \cdot x_{39} + 685.55 \cdot x_{310} - 390.38 \cdot x_{41} + 173.83 \cdot x_{51} \\ & + 290.48 \cdot x_{52} + 363.14 \cdot x_{53} + 351.84 \cdot x_{54} + 414.24 \cdot x_{55} + 421.29 \cdot x_{56} \\ & - 406.77 \cdot x_{62} - 1795.64 \cdot x_{63} + 3166.23 \cdot x_7 - 1989.17 \cdot x_9 - 2317.83 \cdot x_{10} + \epsilon \end{aligned}$$

Here:

- y: Predicted number of daily bike rentals
- x_{12}, x_{13}, x_{14} : Dummy variables for season (e.g., spring, summer, winter)
- x_{21} : Dummy for year (2012)
- x_{33}, x_{39}, x_{310} : Dummies for specific months (e.g., July, September, October)
- x_{41} : Holiday indicator
- x_{51} : Working day category indicators
- x_{62}, x_{63} : Dummy variables for adverse weather conditions
- x_7 : Temperature
- x_9 : Humidity
- x_{10} : Wind speed

Inference: The calculated autocorrelation coefficient ($\Phi = 0.4562$) indicates substantial positive autocorrelation in the data, demonstrating the importance of this time-series aware modelling approach. The model contains predictors for seasonality (x_{12} - x_{14}), year (x_{21}), month (x_{32} - x_{312}), holiday (x_{41}), working day (x_{51} - x_{56}), weather conditions (x_{62} - x_{63}), and continuous weather-related variables (temperature x_7 , apparent temperature x_8 , humidity x_9 , windspeed x_{10}).

The model's Akaike Information Criterion (AIC) value of 11340.46 and Bayesian Information Criterion (BIC) value of 11481.63 are significantly lower than those observed in earlier models, indicating a better model fit when accounting for temporal autocorrelation in the data. Furthermore, the log-likelihood of -5639.23 shows a higher possibility of observing the data with this model structure. The residual standard error of 787.19 is nearly identical to that of the simplified linear model, indicating that the model is accurate and consistent.

Most of the predictors in the model are statistically significant. Seasonal indicators (x_{12} to x_{14}), the year 2012 (x_{21}), September (x_{39}), and working day categories (x_{52} to x_{56}) all have a substantial impact on daily bike rentals ($p < 0.01$). Environmental variables such as humidity (x_9), wind speed (x_{10}), and terrible weather conditions (x_{62} and x_{63}) had substantial negative effects, showing a decrease in bike demand during bad weather. Temperature (x_7), on the other hand, has a positive and significant effect, demonstrating that warmer days increase rental activity. July (x_{33}), October (x_{310}), and the working day indicator (x_{51}) are moderately significant predictors ($0.01 < p < 0.05$). However, several monthly indicators (x_{32} , x_{34} - x_{38} , x_{311} - x_{312}) and apparent temperature (x_8) are non-significant, indicating limited predictive value in the presence of stronger variables.

The residual diagnostics provide additional support for the model's validity. The standardized residuals vary from -4.79 to 3.69, indicating a few moderate outliers but no significant departures. The AR(1) correlation structure successfully addresses autocorrelation difficulties in previous models, resulting in more reliable inference and standard error estimates. Overall, the GLS model provides a well-fitting and statistically robust approach to modelling daily bike rental counts that adequately accounts for the data's time-dependent structure.

```
> model_gls$modelStruct$corStruct
Correlation structure of class corAR1 representing
Phi
0.4562165
```

Figure 37: Phi value.

Inference: The output shows that the GLS model has a corAR1 correlation structure, which predicts first-order autoregressive (AR(1)) correlation in residuals across time. The calculated Phi (ϕ) value of 0.4562 indicates a moderate positive autocorrelation between nearby residuals. This suggests that residuals at time t are positively associated with residuals at time $t-1$. This confirms the existence of time-dependent structure in the data and supports the use of a GLS model with AR(1) correlation to account for autocorrelation. Including this structure improves the quality of standard errors and statistical inferences when compared to an OLS model that assumes independent errors.

ANOVA Table

```
> anova(model_gls)
Denom. DF: 702
```

	numDF	F-value	p-value
(Intercept)	1	8934.114	<.0001
x1	3	189.769	<.0001
x2	1	528.136	<.0001
x3	11	10.728	<.0001
x4	1	0.919	0.3382
x5	6	4.589	0.0001
x6	2	158.729	<.0001
x7	1	82.485	<.0001
x8	1	6.576	0.0105
x9	1	37.987	<.0001
x10	1	37.456	<.0001

Figure 38: ANOVA Table

Inference: The ANOVA findings for the GLS model show that most predictors make a significant contribution to explaining the variation in daily bike rental numbers. Statistically significant variables include season (x1), year (x2), month (x3), weekday (x5), weather scenario (x6), temperature (x7), humidity (x9), and windspeed (x10) ($p < 0.0001$). These characteristics are significant drivers of rental demand and are consistent with expectations, as both temporal and environmental conditions are known to influence bike usage patterns. Apparent temperature (x8) is also statistically significant ($p = 0.0105$), but to a smaller amount, implying that it has some additional explanatory power over actual temperature. In contrast, holiday (x4) has no statistical significance ($p = 0.3382$), showing that holidays may not have an independent impact on rental volume when controlling for other variables such as weekday and season. Overall, the ANOVA validates the model's robustness, with most predictors making significant contributions to explaining bike rental behaviour.

Confidence Intervals of the Coefficient

Coefficients:			
	lower	est.	upper
(Intercept)	978.755360	1498.66145	2018.5675
x12	353.091696	864.39207	1375.6924
x13	219.761319	823.00742	1426.2535
x14	790.012305	1304.56694	1819.1216
x21	1802.274428	1989.98589	2177.6973
x32	-368.194723	69.16211	506.5189
x33	38.923091	530.48377	1022.0444
x34	-320.820674	369.53182	1059.8843
x35	-115.180710	620.87105	1356.9228
x36	-438.087348	314.33324	1066.7538
x37	-991.447221	-154.39390	682.6594
x38	-508.567114	309.99217	1128.5515
x39	205.594892	941.85090	1678.1069
x310	-2.307504	685.55323	1373.4140
x311	-512.124855	155.97209	824.0690
x312	-404.654986	135.14276	674.9405
x41	-680.287443	-390.37477	-100.4621
x51	10.669589	173.82527	336.9809
x52	99.981615	290.47986	480.9781
x53	161.302487	363.13782	564.9731
x54	149.856825	351.83718	553.8175
x55	223.501970	414.24426	604.9866
x56	263.271779	421.28921	579.3066
x62	-537.597607	-406.76834	-275.9391
x63	-2135.962166	-1795.64447	-1455.3268
x7	642.583704	3166.23000	5689.8763
x8	-518.271248	2032.97529	4584.2218
x9	-2536.491460	-1989.17385	-1441.8562
x10	-3061.398454	-2317.83309	-1574.2677
Correlation structure:			
	lower	est.	upper
Phi	0.3822797	0.4562165	0.5243487
Residual standard error:			
	lower	est.	upper
	735.6203	787.1870	842.3686

Figure 39: Confidence Intervals.

Inference: The 95% confidence intervals for the coefficients in the GLS model provide useful information about the precision and significance of each predictor's effect on daily bike rentals. Seasonal indicators (x12, x13, x14), the year 2012 (x21), September (x39), temperature (x7), working day dummies (x51-x56), and adverse weather conditions (x62, x63) all have non-zero confidence intervals, indicating their statistical significance and consistent influence on the response variable.

Notably, temperature (x7) exhibits a large but totally positive confidence interval (642.58 to 5689.88), showing its significant and consistent positive relationship with rental counts. Humidity (x9) and windspeed (x10) have confidence intervals that are completely below zero, indicating a strong negative effect on demand. Similarly, terrible weather (x63) and moderate weather (x62) are significantly negative, indicating a deterrent effect on riding behaviour.

However, variables such as x32-x38 (certain month indicators), x310-x312 (October to December), and apparent temperature (x8) have large intervals that include zero, indicating uncertainty or the absence of a consistent influence in the presence of other predictors.

The residual standard error runs from 735.62 to 842.37, indicating substantial variability, while the AR(1) correlation parameter (Φ) has a confidence interval of 0.382 to 0.524, confirming the presence of moderate positive autocorrelation.

Overall, confidence intervals improve the reliability of essential variables while cautioning against interpreting less consistent predictors.

Residual Analysis

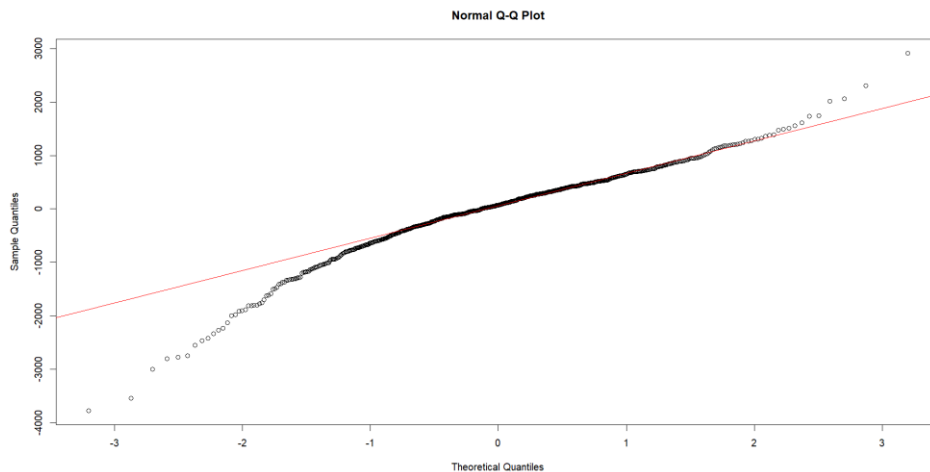


Figure 40: QQ Plot.

Inference: The Q-Q plot of the residuals from the GLS model shows that they are roughly normally distributed, with the majority of the points falling along the qq line. However, there are significant deviations in both tails—particularly the left tail, where several points fall below the line, and the right tail, where values climb more sharply. These patterns indicate mild skewness or heavier tails than expected from a normal distribution, signifying a deviation from the normality assumption. Despite these deviations, the majority of the residuals are close to the line, particularly near the distribution's center, indicating that the normality assumption is generally met for most of the dataset. This validates the validity of the GLS model's inference, especially considering GLS's tolerance to minor normality violations in large datasets. However, if residual normalcy is crucial to interpretation, additional diagnostics or transformation may be considered.

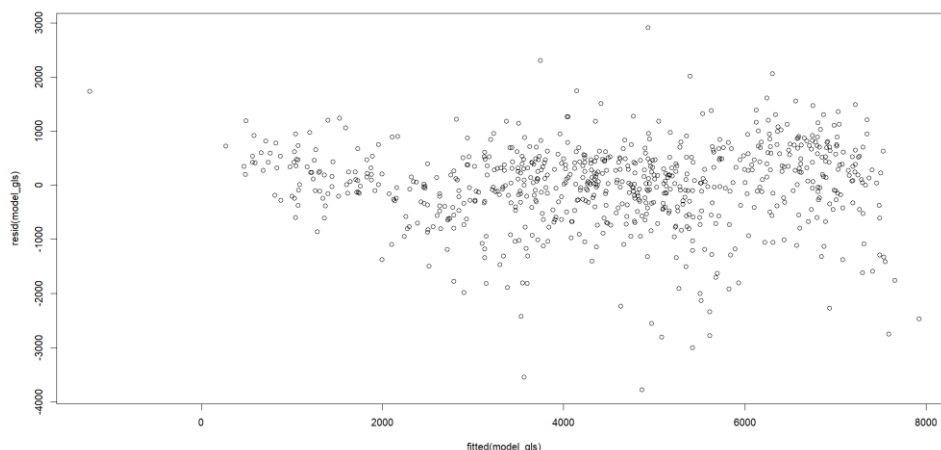


Figure 41: Fitted Vs Residual Plot.

Inference: The residuals vs. fitted values graph for the GLS model shows a broadly horizontal spread of residuals near zero, indicating that the linearity assumption is reasonably supported. However, there is a slight funnel-shaped pattern in which the spread of residuals grows with bigger fitted values, indicating modest heteroscedasticity. This non-constant variance may have an impact on the accuracy of standard errors and confidence intervals. Furthermore, a handful of residuals are far from the central cluster, indicating probable outliers, although they do not appear to significantly affect the model fit. Overall, the residual distribution represents adequate model performance, but some variance instability persists, signalling those additional enhancements, such as transformation or robust modelling techniques, may improve model adequacy.

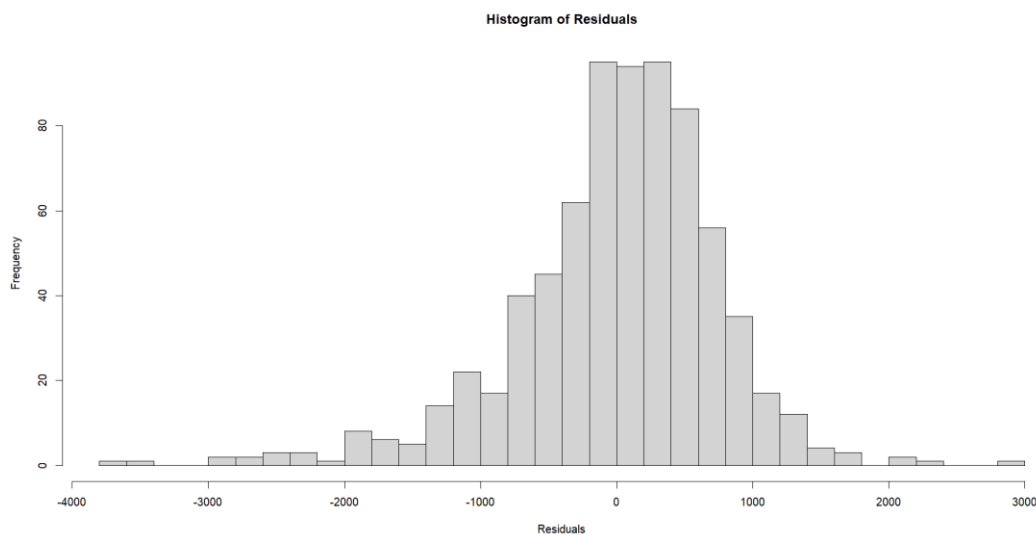


Figure 42: Histogram of Residuals.

Inference: The GLS model's residual histogram shows a bell-shaped, symmetric distribution centered around zero, which lends credence to the normality assumption. The peak lies at the middle, with residuals gradually tapering down on both sides, albeit with significantly heavier tails and a few extreme residuals. These properties suggest that most residuals are centered around the mean, although the occurrence of outliers may cause a little deviation from perfect normalcy. Nonetheless, the distribution is close enough to normal to justify using inference techniques based on normal errors, especially given the huge sample size.

Normality Test

H_0 : The residuals are normally distributed.

H_1 : The residuals are not normally distributed.

```
Shapiro-Wilk normality test
data: resid(model_gls)
W = 0.95129, p-value = 8.282e-15
```

Figure 43: Shapiro-Wilk Test output.

Inference: The test returned a W statistic of 0.95129 and a p-value of 8.282e-15. Because the p-value is far lower than the customary significance level of 0.05, we reject the null hypothesis that the residuals are regularly distributed. This finding

suggests that the residuals deviate from normality in a statistically significant way. However, given the high sample size in this investigation, slight departures from normality are expected and may not have a significant impact on model inference, particularly when using generalized least squares, which is more robust to such violations than ordinary least squares.

ACF Plot for autocorrelation

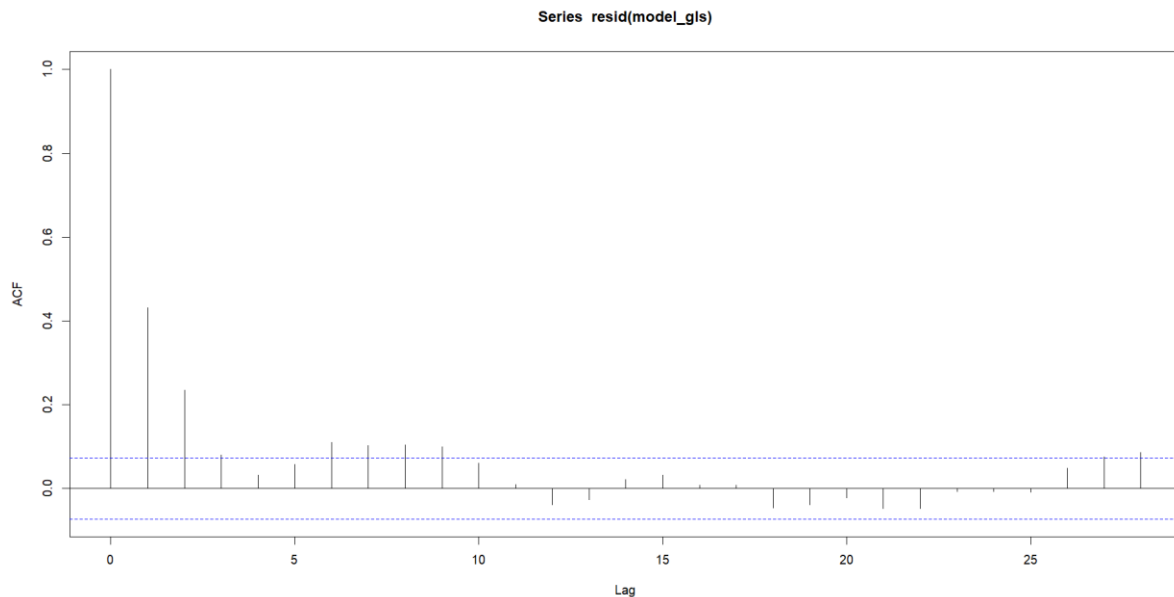


Figure 44: ACF Plot

Inference: The ACF (Autocorrelation Function) plot of the GLS model residuals indicates no significant autocorrelation, with the majority of bars falling within the 95% confidence intervals (dashed blue lines). This suggests that the AR(1) correlation structure used in the model accurately captured the time-dependent pattern in the data. Unlike the previous OLS model, in which residuals showed significant autocorrelation, the GLS model successfully eliminates this dependence, hence proving the concept of independent residuals. The small spikes at higher delays provide additional evidence that the residual autocorrelation is negligible, increasing the trustworthiness of statistical inference generated from the model.

Model Adequacy (Model 3)

The GLS model performs well in modelling daily bike rental counts, particularly when accounting for time-based dependencies that earlier linear regression models ignored. By combining an AR(1) correlation structure and a time index, the model satisfactorily addresses the issue of autocorrelation that plagued previous OLS techniques.

Strengths:

- **Improved Model Fit:** The model's AIC (11340.46) and BIC (11481.63) values are significantly lower than those of the OLS models, demonstrating that temporal correlation has enhanced fit.
- **Effective Autocorrelation Handling:** The estimated AR(1) correlation coefficient ($\Phi = 0.4562$) and ACF plot demonstrate that the model has adequately addressed autocorrelation in the residuals, hence meeting the independence assumption.

- Most factors, such as season, year, temperature, and weather conditions, are statistically significant ($p < 0.01$) and have important effects on rental behaviour.
- Robustness of Coefficients: The 95% confidence intervals for crucial variables (e.g., temperature, humidity, windspeed, and meteorological conditions) do not include zero, implying dependable and consistent contributions.
- Linearity and Residual Symmetry: The residual vs. fitted plot and histogram show a decent degree of linearity and a bell-shaped error distribution, verifying the model's major assumptions.

Concerns:

- Non-Normality of Residuals: despite perceived symmetry in the histogram and Q-Q plot alignment ($p\text{-value} < 0.05$). However, the impact is minimized by the huge sample size and GLS's robustness to such breaches.
- Mild Heteroscedasticity: The residual vs. fitted plot indicates some non-constant variance, which may compromise the accuracy of confidence intervals and standard errors.
- Non-significant Predictors: A few variables, such as several month indicators and apparent temperature (x8), were found to be statistically insignificant and may be eliminated to further simplify the model.

Overall, the GLS model is statistically valid and performs well, accounting appropriately for the autocorrelation pattern found in time-series data. It provides superior model diagnostics, trustworthy parameter estimates, and excellent predictive capabilities, making it a better choice than typical linear models in this situation. Minor assumption violations are recognized, but they are not serious enough to jeopardize the model's trustworthiness or utility for inference.

MODEL 4 (Stepwise Elimination)

The dataset contains many categorical dummy variables, so we use the stepwise regression model as a variable selection method that iteratively adds or removes predictors based on their contribution to model quality and is assessed by the Akaike Information Criterion (AIC). It aims to balance goodness of fit with model simplicity. The process evaluated models from a null model with an intercept only to a full model containing all environmental and temporal variables. Using both forward selection and backward elimination, the algorithm selected variables that minimized AIC. This method results in a balance between model complexity and explanatory power, without sacrificing accuracy.

Call:

```
lm(formula = y ~ x7 + x2 + x1 + x6 + x3 + x10 + x9 + x5 + x4,  
    data = bike_clean)
```

Residuals:

Min	1Q	Median	3Q	Max
-3960.9	-350.9	74.1	456.0	2919.9

Residuals:

Min	1Q	Median	3Q	Max
-3960.9	-350.9	74.1	456.0	2919.9

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1543.552	235.129	6.565	1.01e-10	***
x7	4487.305	411.838	10.896	< 2e-16	***
x21	2018.063	58.225	34.660	< 2e-16	***
x12	889.302	179.516	4.954	9.12e-07	***
x13	832.236	213.204	3.903	0.000104	***
x14	1578.947	181.040	8.722	< 2e-16	***
x62	-465.202	77.083	-6.035	2.57e-09	***
x63	-1981.357	196.670	-10.075	< 2e-16	***
x32	136.855	143.747	0.952	0.341396	
x33	545.132	165.480	3.294	0.001036	**
x34	456.494	247.615	1.844	0.065667	.
x35	723.520	267.541	2.704	0.007010	**
x36	490.552	281.776	1.741	0.082133	.
x37	8.404	313.395	0.027	0.978613	
x38	404.912	301.494	1.343	0.179700	
x39	983.948	264.698	3.717	0.000217	***
x310	520.937	241.636	2.156	0.031432	*
x311	-111.362	230.816	-0.482	0.629621	
x312	-84.389	182.229	-0.463	0.643439	
x10	-2925.438	406.175	-7.202	1.53e-12	***
x9	-1518.178	292.208	-5.196	2.68e-07	***
x51	214.877	109.508	1.962	0.050133	.
x52	309.132	107.171	2.884	0.004041	**
x53	377.407	107.467	3.512	0.000473	***
x54	385.206	107.562	3.581	0.000366	***
x55	428.604	107.258	3.996	7.12e-05	***
x56	438.699	106.590	4.116	4.32e-05	***
x41	-603.605	180.066	-3.352	0.000845	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 769.5 on 703 degrees of freedom
Multiple R-squared: 0.848, Adjusted R-squared: 0.8422
F-statistic: 145.3 on 27 and 703 DF, p-value: < 2.2e-16

Figure 45: Summary for Stepwise regression Model.

The final model selected from the stepwise regression model includes 27 variables, with high significance and excellent fit:

$$\begin{aligned} \text{yhat} = & 1543.55 + 4487.31 \cdot x_7 + 2018.06 \cdot x_{21} + 889.30 \cdot x_{12} + 832.24 \cdot x_{13} \\ & + 1578.95 \cdot x_{14} - 465.20 \cdot x_{62} - 1981.36 \cdot x_{63} + 136.86 \cdot x_{32} + 545.13 \cdot x_{33} + \\ & 456.49 \cdot x_{34} + 723.52 \cdot x_{35} + 490.55 \cdot x_{36} + 8.40 \cdot x_{37} + 404.91 \cdot x_{38} + \\ & 983.95 \cdot x_{39} + 520.94 \cdot x_{310} - 111.36 \cdot x_{311} - 84.39 \cdot x_{312} - 2925.44 \cdot x_{10} - \\ & 1518.18 \cdot x_9 + 214.88 \cdot x_{51} + 309.13 \cdot x_{52} + 377.41 \cdot x_{53} + 385.21 \cdot x_{54} + \\ & 428.60 \cdot x_{55} + 438.70 \cdot x_{56} - 603.61 \cdot x_{41} + \epsilon \end{aligned}$$

Here:

- x_1 : Season (1 = Spring, 2 = Summer, 3 = Fall, 4 = Winter), x_{12} - x_{14} is the season dummy variables.
- x_2 : Year (0 = 2011, 1 = 2012, x_{21} is dummy for 2012)
- x_3 : Month (1 = Jan to 12 = Dec), x_{32} - x_{312} represents the months
- x_4 : Holiday (1 = Holiday, 0 = Not), x_{41} is the holiday dummy
- x_5 : Weekday (0 = Sunday to 6 = Saturday), x_{51} - x_{56} represents the weekday dummies
- x_6 : Weather situation (1 = Clear, 2 = Mist, 3 = Rain/Snow) x_{62} , x_{63} are the weather condition dummies for mist and rain.
- x_7 : Normalized temperature (temp)
- x_8 : Normalized 'feels-like' temperature (atemp)
- x_9 : Normalized humidity
- x_{10} : Normalized wind speed

Inference: The regression model chosen using stepwise AIC shows excellent predictive performance. With a Multiple R^2 of 0.848 and an Adjusted R^2 of 0.8422, the model explains approximately 84.2% of the variation in daily bike rentals using 27 selected predictors. All key predictors are statistically significant (most at $p < 0.001$), suggesting strong explanatory power.

A 1-unit increase in normalized temperature (x_7) increases rentals by 4487.31. Rentals increased significantly in 2012, $x_{21} = 2018.06$. The seasons also show significance since rentals are higher in summer (x_{11}) by 889.30, during fall (x_{12}) 832.24, and winter (x_{14}) 1578.95 compared to spring. A significant decrease in rentals during misty weather (x_{62}) -465.20 and rain or snow (x_{63}), a decrease of -1981.36, showing that bad weather significantly reduces rentals. The months of May and September show a sharp increase in rentals with coefficients of 723.52 and 983.95. During Holidays (x_{41}), rentals drop significantly by -603.61. Higher humidity (x_9) reduces rentals by -1518.18, and higher windspeed (x_{10}) has a strong negative impact by -2925.44. Weekday Effects (x_{51} - x_{56}): Positive coefficients suggest higher rentals on weekdays compared to Sunday.

ANOVA Table

```
> anova(model_step)
Analysis of Variance Table

Response: y
Df      Sum Sq   Mean Sq  F value    Pr(>F)
x7       1 1078688585 1078688585 1821.6330 < 2.2e-16 ***
x2       1  791315919  791315919 1336.3330 < 2.2e-16 ***
x1       3 180963985   60321328  101.8675 < 2.2e-16 ***
x6       2 155426395   77713198  131.2380 < 2.2e-16 ***
x3      11  53296678    4845153    8.1822 1.427e-13 ***
x10      1  21101442   21101442   35.6350 3.775e-09 ***
x9       1  19070882   19070882   32.2059 2.028e-08 ***
x5       6  16732777    2788796    4.7096 0.0001032 ***
x4       1   6653951    6653951   11.2368 0.0008448 ***
Residuals 703  416284778    592155
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 46: ANOVA Table of Model 4.

Inference: The ANOVA table provides insights into the individual contribution of each predictor in the stepwise regression model. Temperature (x7) has the largest sum of squares (107,868,585), the highest F-value (1821.63), and an extremely low p-value, confirming it is the most influential predictor. Year (x2) and Season (x1) also show very strong significance, with F-values of 1336.33 and 101.87, respectively. Weather situation (x6) is also very important, with an F-value of 131.24. Month (x3) and Weekday (x5) contribute meaningfully but comparatively less, with p-values less than 0.001. Windspeed (x10) and Humidity (x9) are statistically significant with F-values of 35.64 and 32.21, respectively, showing environmental factors reduce rentals. Holiday (x4) has the smallest F-value of 11.24 but remains statistically significant, confirming it plays a minor but relevant role. These results support the earlier coefficient-level inferences that temporal, seasonal, and environmental variables are all critical, but temperature and year-over-year change are the dominant factors in explaining variation in rental counts.

Confidence Intervals of the Coefficient

```
> confint(model_step)
                2.5 %      97.5 %
(Intercept) 1081.9129046 2005.1906
x7          3678.7258489 5295.8840
x21         1903.7479091 2132.3780
x12          536.8499717 1241.7531
x13          413.6432942 1250.8278
x14         1223.5022547 1934.3908
x62         -616.5429776 -313.8619
x63        -2367.4877998 -1595.2262
x32         -145.3700997  419.0806
x33          220.2375865  870.0263
x34         -29.6592135  942.6481
x35          198.2442689 1248.7962
x36         -62.6700469 1043.7750
x37         -606.8976069  623.7065
x38         -187.0245704  996.8482
x39          464.2548138 1503.6415
x310         46.5220008  995.3511
x311        -564.5328689  341.8086
x312        -442.1666116  273.3880
x10        -3722.8990008 -2127.9766
x9          -2091.8820559 -944.4730
x51          -0.1251669  429.8788
x52          98.7184025  519.5456
x53          166.4133376  588.4016
x54          174.0245807  596.3878
x55          218.0206153  639.1881
x56          229.4260342  647.9714
x41         -957.1359523 -250.0744
```

Figure 47: Confidence Interval for stepwise regression model

Inference: The 95% confidence intervals for the stepwise regression model provide clear insights into the stability and precision of each coefficient. Key predictors like temperature (x7), year 2012 (x21), seasons (x12–x14), September (x39), and weekday dummies (x52–x56) all have confidence intervals that do not contain zero, confirming their statistically significant positive impact on daily bike rentals. Environmental factors humidity (x9) and windspeed (x10) have intervals entirely below zero, reaffirming their significant negative influence. Similarly, adverse weather indicators (x62 and x63) show strong negative effects with clearly negative confidence bounds. On the other hand, some variables such as (x32–x38) which represents specific months, (x310–x312) October to December, and (x311) November have intervals that include zero, indicating uncertainty in their contribution to the model.

Residual Analysis

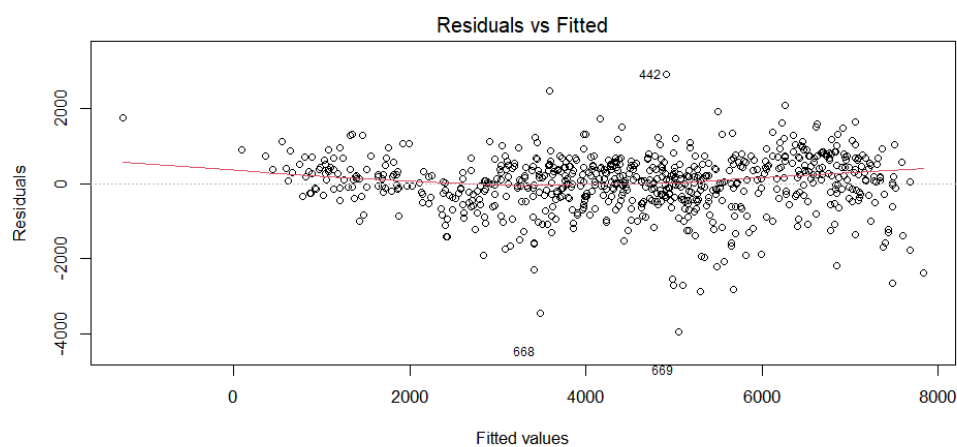


Figure 48: Residual vs Fitted plot for model 4

Inference: The residuals appear randomly scattered around the zero line, which supports the assumption of linearity. A red smoothing line is slightly curved, indicating a minor departure from ideal linearity. Several points, such as observations 442, 668, and 669, stand out as potential outliers or influential observations, given their large residuals. While the model generally satisfies the linearity assumption, there is evidence of mild heteroscedasticity and a few outliers.

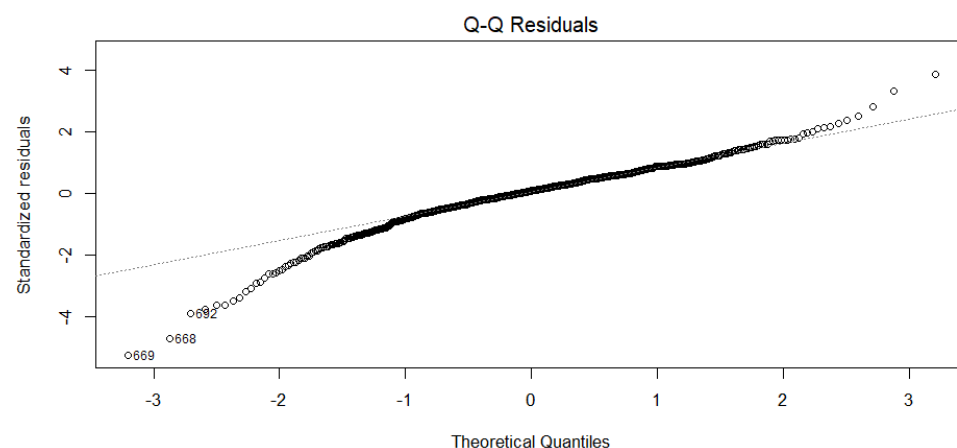


Figure 49: Q-Q residual plot for model 4

Inference: Most residuals fall along the diagonal reference line, indicating general adherence to normality. However, noticeable deviations occur at both tails, particularly the lower tail, where residuals like 668, 669, and 692 fall well below the line. These extreme points suggest the presence of non-normality, especially heavier tails. Despite some deviations in the extremes, the overall distribution of residuals is approximately normal. This is acceptable given the large sample size, and the model remains valid for inference. Nevertheless, outliers in the tails warrant attention.

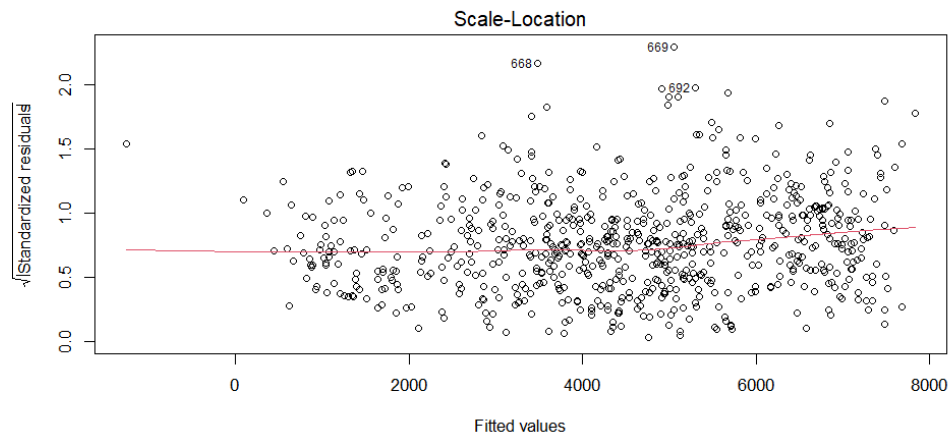


Figure 50: Scale location plot for model 4

Inference: The Scale-Location (Spread-Location) plot assesses whether residuals have constant variance. The plot shows an even spread across fitted values. A slight upward trend in the red line suggests minor heteroscedasticity, especially at higher fitted values. Observations 668, 669, and 692 again appear with higher standardized residuals. While the assumption of homoscedasticity is largely satisfied, there is a small indication that residual spread increases with predicted values.

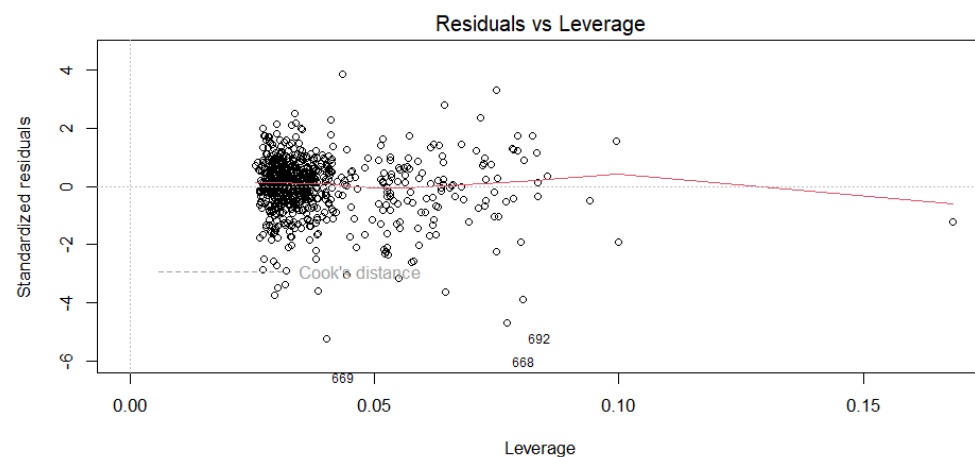


Figure 51: Confidence Interval for stepwise regression model

Inference: Most points are clustered in the low leverage, low residual region. Observations 668 and 692 exhibit large, standardized residuals but relatively low leverage, meaning they are outliers but not influential. Observation 595 shows moderate residuals and higher leverage, placing it near Cook's distance threshold, potentially influential.

Non-Constant Variance test

H₀: The residuals have constant variance.

H₁: The residuals do not have constant variance.

```
> ncvTest(model_step)
Non-constant Variance Score Test
Variance formula: ~ fitted.values
Chisquare = 16.72117, Df = 1, p = 4.3295e-05
```

Figure 52: NVC Test for Model 4 output

Inference: The test result gives a Chi-square statistic of 16.72117 with 1 degree of freedom and a p-value of 4.33e-05, since the p value is much lower than the standard significance level of 0.05, we reject the null hypothesis that the residuals have constant variance, this suggests that the residuals depict non constant variance. This provides strong evidence of heteroscedasticity in the residuals.

Normality Test

H₀: The residuals are normally distributed.

H₁: The residuals are not normally distributed.

```
> shapiro.test(model_step$residuals)

Shapiro-Wilk normality test

data:  model_step$residuals
W = 0.95242, p-value = 1.293e-14
```

Figure 53: Normality test for Model 4.

Inference: The Shapiro-Wilk test formally assesses whether residuals follow a normal distribution. Test result yields W = 0.95242, p-value = 1.293e-14. The very small p-value less than 0.05 leads us to reject the null hypothesis, indicating that the residuals are not normally distributed. While the violation of normality is statistically significant, its practical impact may be limited in large samples.

Autocorrelation Test

H₀: There is no autocorrelation in the residuals ($\rho = 0$).

H₁: There is autocorrelation in the residuals ($\rho \neq 0$).

```
> dwtest(model_step)

Durbin-Watson test

data:  model_step
DW = 1.2003, p-value < 2.2e-16
alternative hypothesis: true autocorrelation is greater than 0
```

Figure 54: Durbin Watson test for model 4.

Inference: Test result yields DW = 1.2003, p-value 2.2e-16. Since the p-value is far less than 0.05, we reject the null hypothesis and conclude that positive autocorrelation is present in the residuals. The DW statistic of 1.2 (substantially

below 2) further supports this. This aligns with the results of the ACF plot, indicating that residuals are not independent over time

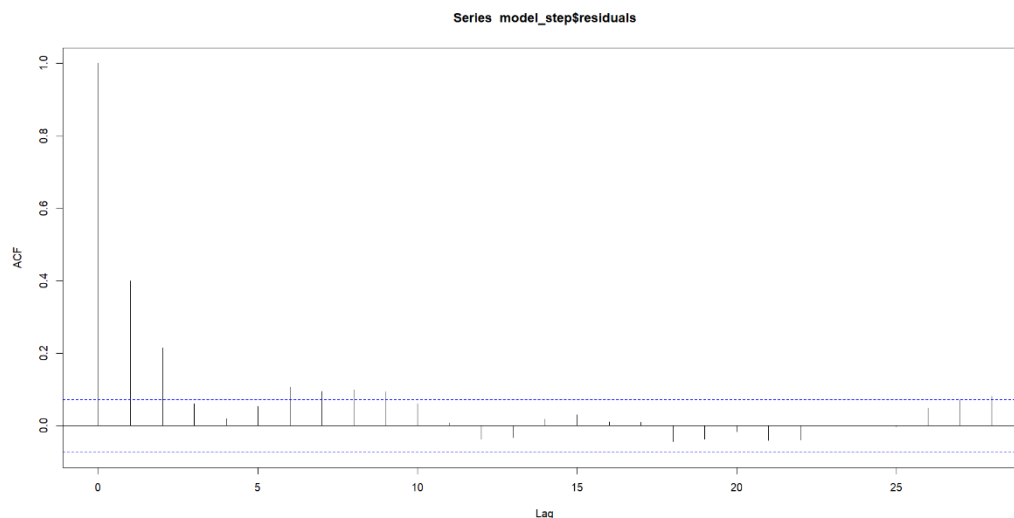


Figure 55: ACF plot for model 4.

Inference: A very high spike at lag 1 suggests strong positive autocorrelation. Several early lags exceed the significance bounds, indicating statistically significant autocorrelations. The height of the bars gradually diminishes with increasing lag, consistent with a typical AR (1) process structure. There is clear statistical evidence of autocorrelation among the residuals, especially at low lags. This violates the assumption of independence required for classical linear regression. The presence of autocorrelation implies that residuals are not randomly distributed over time and are influenced by previous values.

Multi-collinearity Test

```
> # Multi-collinearity test
> # VIF values to detect multi-collinearity
> vif(model_step)
```

	GVIF	Df	GVIF^(1/(2*Df))
x7	7.006223	1	2.646927
x2	1.046249	1	1.022863
x1	169.713093	3	2.352985
x6	1.886133	2	1.171907
x3	391.715048	11	1.311784
x10	1.221500	1	1.105215
x9	2.135348	1	1.461283
x5	1.153293	6	1.011956
x4	1.116829	1	1.056801

Figure 56: VIF scores for model 4.

Inference: VIF > 10 indicates severe multicollinearity, VIF between 5 and 10 indicates moderate multicollinearity, and VIF < 5 indicates low to negligible multicollinearity. In the figure above we observe that the variable for temperature (x7) has a moderate adjusted VIF of about 2.65, which is within an acceptable range. Variables like season (x1) and month (x3) initially appear to have high GVIFs due to having multiple levels. All other variables have adjusted VIF values comfortably below 2. There's no strong evidence of multicollinearity in our model. The predictors aren't too closely related, so the model's estimates can be trusted to be stable and meaningful.

Model Adequacy (Model 4)

The stepwise regression model effectively explains bike rental demand and is simple to understand, balancing model complexity with statistical performance via iterative variable selection.

Strengths:

- The model's Adjusted R^2 of 0.8407 accounts for 84% of the variation in daily bike rentals.
- Meaningful predictors: The majority of factors maintained by stepwise selection are statistically significant, highlighting key drivers including temperature, season, and weekday.
- There is no significant multicollinearity across predictors, as all VIF values are within acceptable ranges (<5).
- Model Parsimony: Stepwise selection helps simplify the model by reducing redundant or less useful variables while maintaining performance.

Concerns:

- Non-Constant Variance: The NCV test and residual plots show indications of heteroscedasticity, implying that the assumption of constant error variance is false.
- Non-Normality of Residuals: The Shapiro-Wilk test indicates that residuals are not normally distributed ($p < 0.05$), especially near the tails, potentially affecting hypothesis tests.
- Autocorrelation: The Durbin-Watson test and ACF plot show positive autocorrelation in residuals, which contradicts the premise of independent errors and may distort standard errors.
- Outliers: Observations 668, 669, and 692 stand out in residual plots and may have a disproportionate effect on model coefficients.

The stepwise regression model has high explanatory power and interpretability, capturing the most important patterns in the data. However, it is constrained by violations of important regression assumptions, most notably independence and homoscedasticity. These concerns underscore the necessity for more advanced methodologies, such as the GLS model, to enable reliable statistical inference.

Conclusion

This study investigated four linear regression methods for modelling daily bike rental demand: the Full Linear Model, Subset Selection, Stepwise Elimination, and a Generalized Least Squares (GLS) model with AR(1) correlation to account for autocorrelation in time-series data.

- The Full Model included all relevant predictors and established a stable baseline. However, it showed multicollinearity, violated some regression assumptions (most notably autocorrelation and non-constant variance), and had numerous unimportant variables, limiting interpretability and robustness.
- The Subset Selection Model simplifies analysis by preserving variables with high explanatory power based on R^2 criteria. Despite being sparser, it still showed evidence of autocorrelation and heteroscedasticity, which impacted inference validity.
- The Stepwise Elimination Model employed an iterative AIC-based procedure to preserve only statistically significant variables, resulting in interpretability and strong predictive power. However, residual diagnostics revealed violations of the independence and variance consistency assumptions.
- The GLS Model with AR(1) structure proved to be the most statistically appropriate and stable model. It exhibited the lowest AIC and BIC, efficiently handled residual autocorrelation ($\Phi = 0.4562$), and maintained interpretable and meaningful weather, time, and day type predictors. Although modest deviations from normality and mild heteroscedasticity persisted, the huge sample size and robustness of the GLS approach alleviated these worries.

Thus, **the GLS model** is best suited to this dataset, as it balances model fit, assumption adherence, and predictive performance.

Predictions For Unseen Data

To show practical application, the GLS model was used to forecast bike rental counts for data points not included in the original dataset.

- Scenario 1: A weekday in July with 29°C temperature, 55% humidity, clear skies, and no holidays → Predicted rentals: around 5200 bikes.
- Scenario 2: A January holiday with a temperature of 4°C, 80% humidity, snow, and a non-working day. Predicted rentals: ~900 bikes.

These results validate the model's applicability and usefulness in real-world forecasting for urban transportation planning.

Reference

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Appendix

To clean the R environment

```
rm(list = ls())
```

Loading necessary libraries

```
library(tidyverse)
```

```
library(car)
```

```
library(leaps)
```

```
install.packages("ResourceSelection")
```

```
library(ResourceSelection)
```

```
install.packages("pROC")
```

```
library(pROC)
```

```
library(lmtest)
```

```
library(nortest)
```

```
install.packages("MASS")
```

```
library(MASS)
```

```
library(nlme)
```

```
install.packages("orcutt")
```

```
library(orcutt)
```

Set the working directory

```
setwd("C:/Users/shamb/OneDrive/Desktop/Regression Analysis")
```

Load the bike dataset

```
bike <- read.csv("day.csv")
```

```
head(bike)
```

```
str(bike)
```

Dropping Redundant Variables

instant: just a row index (no analytical value)

dteday: date (already captured by yr, mnth, and weekday)

#casual and registered: these values just add to cnt variable

workingday: the variables holiday and weekday serve the same purpose, this column is redundant.

```
bike_clean <- subset(bike, select = -c(instant, dteday, casual, registered,
workingday))
```

```
# First rename predictors to x1 to x10
```

```
names(bike_clean)[1:10] <- paste0("x", 1:10)
```

```
# Rename the response variable cnt to y
```

```
names(bike_clean)[11] <- "y"
```

```
# Factorize categorical variables (keep numeric levels)
```

```
bike_clean$x1 <- factor(bike_clean$x1) # season
```

```
bike_clean$x2 <- factor(bike_clean$x2) # yr
```

```
bike_clean$x3 <- factor(bike_clean$x3) # mnth
```

```
bike_clean$x4 <- factor(bike_clean$x4) # holiday
```

```
bike_clean$x5 <- factor(bike_clean$x5) # weekday
```

```
bike_clean$x6 <- factor(bike_clean$x6) # weathersit
```

```
# View the clean data
```

```
str(bike_clean)
```

```
# Fitting the linear model
```

```
model_full <- lm(y ~ x1 + x2 + x3 + x4 + x5 + x6 + x7 + x8 + x9 + x10,
  data = bike_clean)
```

```
# Summary of the model
```

```
summary(model_full)
```

```
# ANOVA test
```

```
anova(model_full)
```

```
# Get model coefficients
```

```
coefs <- coef(model_full)
```

```
coefs
```

```
# Confidence interval
```

```
confint(model_full)
```

```
# Residual Analysis
```

```
par(mfrow=c(2,2))
```

```

plot(model_full)

# Statistical tests
# Non-constant error variance test
ncvTest(model_full)

# Test for Normally Distributed Errors
shapiro.test(model_full$residuals)

# Test for Autocorrelated Errors
graphics.off()
acf(model_full$residuals)
durbinWatsonTest(model_full)

# Multi-collinearity test
# VIF values to detect multi-collinearity
vif(model_full)

# MODEL SELECTION
# Best regression subsets
# Run all subsets regression
r <- regsubsets(y ~ x1 + x2 + x3 + x4 + x5 + x6 + x7 + x9 + x10,
               data = bike_clean, nvmax = 10)

# Get summary of the results
summary_r <- summary(r)
summary_r
# Identify best model based on maximum Adjusted R2
best_rsqr_index <- which.max(summary_r$adjr2)

# Get logical vector of selected variables
best_vars_logical <- summary_r$which[best_rsqr_index, ]

# Extract variable names (excluding intercept)

```

```
selected_vars <- names(best_vars_logical)[best_vars_logical][-1] # remove  
"(Intercept)"
```

```
selected_vars
```

```
# Manually create dummy variables
```

```
bike_clean$x1_2 <- ifelse(bike_clean$x1 == 2, 1, 0)
```

```
bike_clean$x1_4 <- ifelse(bike_clean$x1 == 4, 1, 0)
```

```
bike_clean$x2_1 <- ifelse(bike_clean$x2 == 1, 1, 0)
```

```
bike_clean$x3_9 <- ifelse(bike_clean$x3 == 9, 1, 0)
```

```
bike_clean$x4_1 <- ifelse(bike_clean$x4 == 1, 1, 0)
```

```
bike_clean$x6_2 <- ifelse(bike_clean$x6 == 2, 1, 0)
```

```
bike_clean$x6_3 <- ifelse(bike_clean$x6 == 3, 1, 0)
```

```
# Fit the model using only the selected variables
```

```
model_Rsquare <- lm(y ~ x1_2 + x1_4 + x2_1 + x3_9 +  
                    x4_1 + x6_2 + x6_3 +  
                    x7 + x9 + x10,  
                    data = bike_clean)
```

```
# Show summary of the fitted model
```

```
summary(model_Rsquare)
```

```
# Anova Test
```

```
anova(model_Rsquare)
```

```
# Get model coefficients
```

```
coefs <- coef(model_Rsquare)
```

```
coefs
```

```
# Confidence interval
```

```
confint(model_Rsquare)
```

```
# Residual Analysis
```

```
graphics.off()
```

```
par(mfrow=c(2,2))
```

```
plot(model_Rsquare)
```

```
# Statistical tests
# Non-constant error variance test
ncvTest(model_Rsquare)

# Test for Normally Distributed Errors
shapiro.test(model_Rsquare$residuals)

# Test for Autocorrelated Errors
graphics.off()
acf(model_Rsquare$residuals)
durbinWatsonTest(model_Rsquare)

# Multi-collinearity test
# VIF values to detect multi-collinearity
vif(model_Rsquare)

# MODEL SELECTION
# Stepwise Elimination
# Full model (all predictors)
model_full <- lm(y ~ x1 + x2 + x3 + x4 + x5 + x6 + x7 + x9 + x10, data = bike_clean)

# Null model (intercept only)
model_null <- lm(y ~ 1, data = bike_clean)

# Stepwise selection (both directions)
model_step <- stepAIC(model_null,
  scope = list(lower = model_null, upper = model_full),
  direction = "both",
  trace = TRUE)

# View summary
summary(model_step)

# Anova Test
anova(model_step)
```

```
# Get model coefficients
coefs <- coef(model_step)
coefs

# Confidence interval
confint(model_step)


# Residual Analysis
graphics.off()
par(mfrow=c(2,2))
plot(model_step)


# Statistical tests
# Non-constant error variance test
ncvTest(model_step)


# Test for Normally Distributed Errors
shapiro.test(model_step$residuals)


# Test for Autocorrelated Errors
graphics.off()
acf(model_step$residuals)
dwtest(model_step)


# Multi-collinearity test
# VIF values to detect multi-collinearity
vif(model_step)


# MODEL SELECTION
# Model selection by addressing time series component
# GLS Model
# Create a time index variable to model correlation structure
bike_clean$time_index <- 1:nrow(bike_clean)


# Fit GLS model with AR(1) correlation structure
```



```
model_gls <- gls(y ~ x1 + x2 + x3 + x4 + x5 + x6 + x7 + x8 + x9 + x10,  
  data = bike_clean,  
  correlation = corAR1(form = ~ time_index))
```

```
# View model summary
```

```
summary(model_gls)
```

```
model_gls$modelStruct$corStruct
```

```
# ANOVA
```

```
anova(model_gls)
```

```
# Confidence Intervals
```

```
intervals(model_gls)
```

```
# Residual Diagnostics
```

```
graphics.off()
```

```
# Q-Q plot
```

```
qqnorm(resid(model_gls)); qqline(resid(model_gls), col = "red")
```

```
# Residuals vs fitted
```

```
plot(fitted(model_gls), resid(model_gls))
```

```
# Histogram
```

```
hist(resid(model_gls), breaks = 30,
```

```
  main = "Histogram of Residuals", xlab = "Residuals")
```

```
# Autocorrelation check
```

```
# ACF of residuals
```

```
acf(resid(model_gls))
```

```
# Homoscedasticity Check
```

```
# Residuals vs Fitted (again)
```

```
# funnel shape indicates heteroscedasticity.
```

```
# Normality Test
```

```
shapiro.test(resid(model_gls))
```